
Marking the Wrong Symptoms: Evaluating LLM Watermarks in Medical Texts

Melanie Rieff¹ Robin Staab¹ Thibaud Gloaguen¹ Stefan Hegselmann^{2,3} Martin Vechev¹

Abstract

As LLMs are increasingly integrated into clinical workflows, watermarking is becoming essential for traceability—yet existing watermarks are evaluated only on general-purpose benchmarks, leaving medicine, where small token-level perturbations can alter clinical meaning, under-explored. We present the first systematic study of how watermarking affects medical performance, evaluating 5 schemes across 11 LLMs and 7 VLMs on unimodal and multimodal clinical reasoning tasks, complemented by a human-expert-validated pipeline auditing reasoning quality, terminological precision, and hallucinations. Our results show that watermarking can induce lexical corruption, hallucinated terminology, and degraded image interpretation—failures that aggregate metrics systematically obscure, establishing domain-specific evaluation as a prerequisite for safe deployment of watermarked models in medicine.

1. Introduction

Large language models (LLMs) are increasingly integrated into clinical workflows, with 81% of U.S. physicians reporting professional AI use [3] and active deployment in report generation, decision support, and patient communication [13, 35]. In parallel, emerging regulation [9] is pushing toward provenance marking of AI-generated content, with watermarking emerging as the leading compliance mechanism [8], meaning clinically deployed LLMs are increasingly likely to be watermarked.

LLM Watermarking Watermarking embeds a statistically detectable signal by modifying the LLM sampling

procedure at every generation step, directly influencing the model’s output distribution. Prior evaluations rely on general-domain benchmarks and coarse proxies such as perplexity. In medicine, where correctness depends on precise reasoning, terminology, qualifiers, and numerical values, even small token-level perturbations can alter clinical meaning, suggesting these results may not transfer [34, 28, 16].

This work We propose the first systematic study of how text watermarking affects medical LLM performance (Figure 1), evaluating 5 prominent watermarks on two medical benchmarks spanning clinical knowledge and visual understanding, with three LLM-as-judges validated against clinical expert annotations (Sec. 3.2). Our results reveal that watermarking can leave benchmark accuracy largely intact while substantially degrading the underlying reasoning: the rate of correct answers backed by flawed reasoning more than doubles on several models (e.g., 11.4% → 26.3% on PHI-4-14B with SynthID), fabricated medical entities increase by up to +39.2 per 100 questions, and on multimodal inputs watermarks corrupt visual interpretation. For reasoning models, watermarking the trace inflates output length by up to 69% while degrading reasoning quality, whereas restricting the watermark to the final answer preserves both (App. C.2).

Key Contributions Our main contributions are:

- A systematic evaluation of five watermarking schemes across 11 LLMs and 7 VLMs on medical benchmarks (MedQA, MedXpertQA-MM), with *physician-validated* LLM-as-a-judge audits assessing reasoning quality, terminological precision, and hallucinations beyond letter accuracy.
- The first analysis of how text-decoding watermarks interact with multimodal medical input, revealing that watermarking during text generation degrades visual grounding—suppressing correct image-based claims and amplifying contradicted ones—despite stable accuracy.
- A study of reasoning model watermarking, showing that restricting watermarks to the user-visible final answer preserves reasoning quality, while watermarking the thinking degrades terminological precision and induces circular deliberation.

¹Department of Computer Science, ETH Zurich, Switzerland

²Berlin Institute of Health at Charité – Universitätsmedizin Berlin, Center of Digital Health, Berlin, Germany ³Deutsches Herzzentrum der Charité – Medical Heart Center of Charité and German Heart Institute Berlin, Berlin, Germany. Correspondence to: Melanie Rieff <mrieff@student.ethz.ch>.

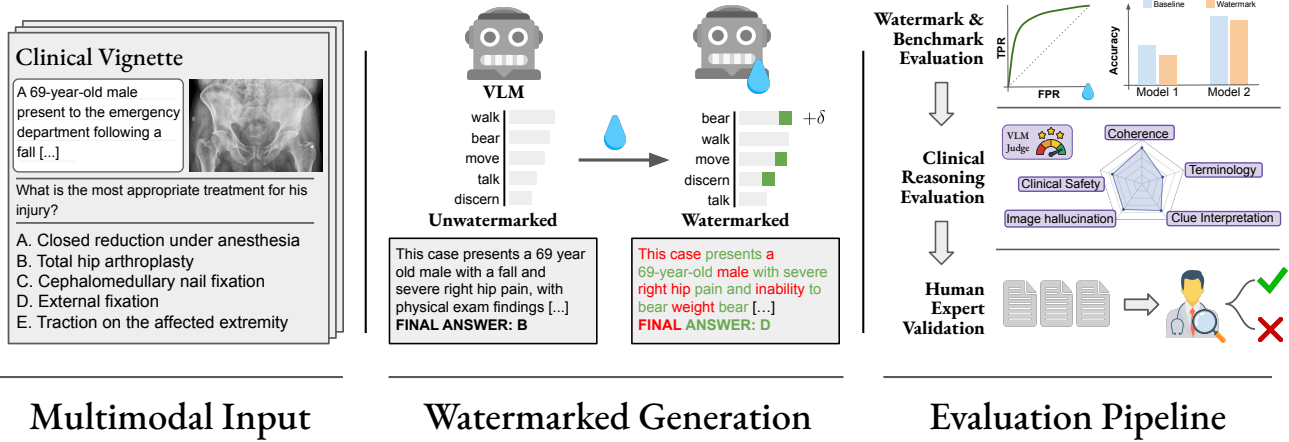


Figure 1. **Overview of our evaluation pipeline.** Given a multimodal benchmark instance, we prompt VLMs (using chain-of-thought prompting) with and without watermarks for reference. We first evaluate the detectability of watermarks in the answers and their accuracy. We then examine the reasoning traces in more depth using three specific LLM-as-judges, validated by clinical experts.

2. Related Work

Medical LLMs are commonly evaluated on USMLE-style [19, 20] or multimodal [18, 6, 40] benchmarks, yet accuracy alone misses flawed reasoning, hallucinations, and terminology misuse [27, 30, 16], motivating richer rubric- and physician-graded protocols [22, 4]. In parallel, generation-time text watermarks [25] embed detectable signals via either *distortionary* reweighting (KGW [21], PPL [11]) or *distortion-free* sampling (DipMark [36], AAR [1], SynthID [7]), and existing evaluations rely on detection power and coarse proxies (perplexity, brief LLM ratings) on general-domain tasks [21, 7, 23], which are poorly aligned with human judgement [34] and mask task-, scale-, and reasoning-dependent degradation [38, 24, 5, 26]. The only prior medical-text watermark study [17] suffers from methodological flaws (no chat templates, 25-token cap, no paired baseline). We close this gap with the first systematic evaluation of five watermarks across multiple model scales, specialisations, and multimodal medical inputs, complemented by physician-validated LLM-as-judges that probe reasoning quality, lexical precision, and diagnostic consistency beyond letter accuracy.

3. Our Evaluation Pipeline

In this section, we introduce our evaluation pipeline illustrated in Figure 1. We summarise the models, watermarks, and data we evaluate (Sec. 3.1), and describe the metrics we use (Sec. 3.2).

3.1. Setup

Our evaluation spans two medical benchmarks (*MedQA* [19], 2,000 four-option USMLE-style questions, and *MedXpertQA-MM* [40], 2,000 five-option,

image-paired questions across 17 specialties), 11 LLMs and 7 VLMs at 4B–70B scale spanning general-purpose families (LLAMA-3.1, GEMMA-3/4, PHI-4, QWEN-3/-VL), reasoning models (DEEPSEEK-R1-DISTILL-LLAMA-8B, LLAMA-70B, QWEN-32B), QWEN-3.5-27B), and medical-specialised checkpoints (OPENBIO-LLM-70B, ULTRAMEDICAL-70B, LINGSHU-7/32B, MEDGEMMA-4/27B), and five prominent generation-time watermarking schemes (KGW [21], DipMark [36], AAR [1], PPL [11], SynthID [7]) covering the major algorithmic families and split into *distortionary* (KGW, PPL, AAR/SynthID at high strength) and *distortion-free* (DipMark, AAR at zero strength, SynthID with binary tournaments) variants. For each scheme, we sweep multiple strength levels to map the detectability–quality trade-off. Together these define the (model, watermark, strength) configurations we evaluate; full benchmark, model, and scheme details, along with generation and reproducibility settings, are deferred to App. A.

3.2. Evaluation Dimensions

We score each (model, watermark, strength) configuration along two fronts. First, we sample full 2,000-question chain-of-thought completions (prompts in App. E.1) and report *accuracy* together with watermark *detectability*, measured as the average true positive rate (TPR) at a 1% false positive rate over the 2,000 answers. Accuracy alone, however, cannot detect fabricated entities, misapplied terminology, or distorted reasoning that lead to correct letters via flawed reasoning [27, 30]; we therefore complement it with three structured LLM-as-judges, using GEMINI-3-FLASH as the judge for its balance of cost and reasoning capability.

1. *Single-response audit*: blind to condition and gold answer, the judge flags eight reasoning/terminology axes

(fabricated entities, corrupted spellings, misapplied real terms, confident error, vignette fabrication, clue misread, within-response contradiction, linguistic corruption, format/termination failure) plus abstention; for MedXpertQA-MM, each perceptual image claim is also rated SUPPORTED/CONTRADICTED/UNVERIFIABLE.

2. *Pairwise divergence audit*: watermarked and unwatermarked completions for the same question are scored side by side (order randomised, labels stripped) for pair correctness, diagnostic divergence (NONE/MINOR/MAJOR), and scenario fabrication.
3. *Reasoning-trace audit* (thinking models): a two-block protocol rates four reasoning-quality dimensions (efficiency, clinical coherence, answer-reasoning alignment, distractor engagement), two factual dimensions (terminological precision, diagnostic accuracy), and two deployment ratings (*clinical safety*, *supervisory burden*); we isolate trace from final-answer degradation via two paired comparisons (full WM vs. FA-only, FA-only vs. no WM; App. C.2).

We validate the judges against physician agreement in Sec. 4.3 and App. D, and run all audits at the lowest watermark strength achieving $\geq 99\%$ TPR at 1% FPR (the operating point prior work considers required for regulatory deployment); full prompts, axis definitions, and decoding configurations are in App. E.2.

4. Results

We evaluate how watermarks affect clinical performance and validate our judges: accuracy remains mostly stable (Sec. 4.1) but reasoning quality degrades sharply (Sec. 4.2); judges are validated in Sec. 4.3 and reasoning-model results in App. C.2.

4.1. Accuracy Looks Mostly Stable, But the Picture is Misleading

For most schemes and models, accuracy stays within the 95% confidence interval (CI) of the unwatermarked baseline (Figure 2; MedQA in Figure 4, appendix). On MedQA, only three configurations drop visibly – the medical-specialised 70Bs (ULTRAMEDICAL-70B, OPENBIOLLM-70B) at high TPR@1%FPR and LLAMA-3.1-8B more broadly – while the equally large general-purpose LLAMA-3.1-70B remains stable, ruling out scale or domain specialisation as sufficient explanations.¹ On MedXpertQA-MM, all VLMs stay within or near the CI, and although perplexity rises monotonically with strength, it does not correlate with ac-

¹BIOMISTRAL-7B also collapses (KGW -37.5 , SynthID -14.2 pp), but is confounded by poor CoT-MCQ instruction tuning (median ~ 6 words per response vs. 250–450); we report it only in appendix grids.

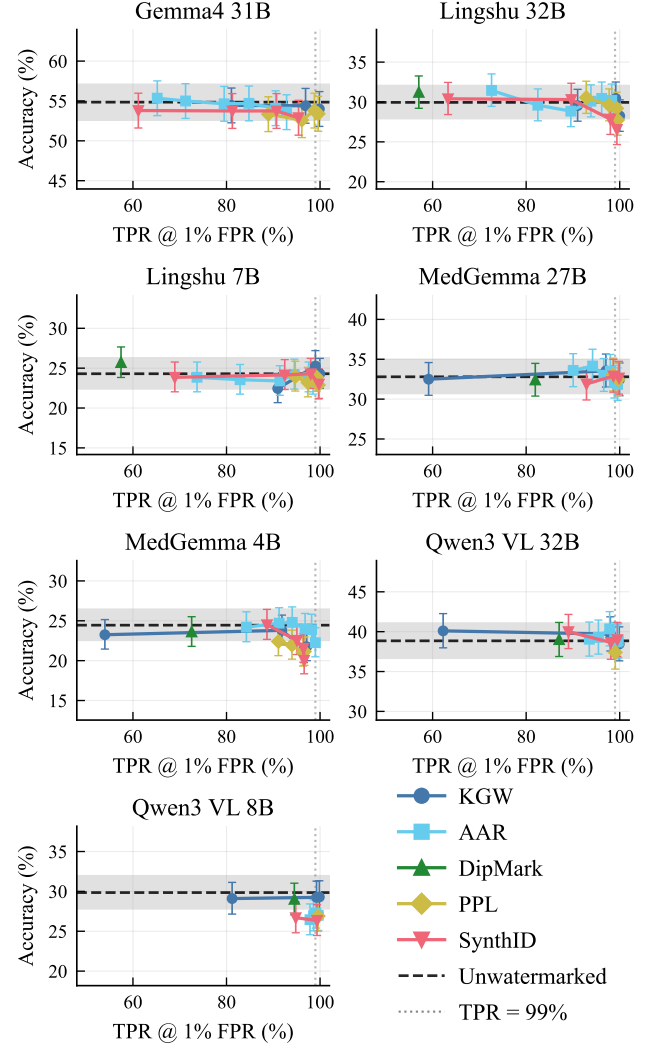


Figure 2. **MedXpertQA-MM accuracy vs. detectability**, seven VLMs. Dashed line and grey band: unwatermarked mean and 95% CI; dotted vertical: TPR = 99%. VLMs consistently stay near the baseline band despite reasoning quality varying noticeably (Sec. 4.2). The corresponding MedQA plot is Figure 4 (appendix).

curacy or reasoning degradation (App. B.1). A standard accuracy benchmark would thus deem watermarking safe for medical use – a conclusion our audits below show is incorrect: even where accuracy holds, watermarking corrupts the reasoning a clinician would actually read.

4.2. Reasoning Quality Degrades Even When Accuracy Holds

We score every (model, scheme) configuration at $\geq 99\%$ TPR using the judge audits of Sec. 3.2, with bootstrap p-values quantifying significance versus unwatermarked. The two most-affected models per benchmark are in Table 2; full results in App. C.1.

Marking the Wrong Symptoms: Evaluating LLM Watermarks in Medical Texts

Type	Watermarked Term	Correct Term / Explanation
Fabricated entity	<i>Streptococcus epidermidis</i>	<i>Staphylococcus epidermidis</i> .
	Pantoprazole	Pantoprazole
	Naftifloxacin	No such antibiotic
	“plutonium channels”	potassium channels (plutonium is a radioactive element)
Misapplied real term	“ β -blockers preferred over CCBs in COPD”	Reversed: β -blockers risk bronchospasm; non-dihydropyridine CCBs are safe in COPD.
	Benzodiazepines “increase duration of GABA-channel opening”	Benzodiazepines increase channel-opening frequency; duration is the mechanism of barbiturates.
Corrupted spelling	<i>Clostridioides diffidile</i> meningitis bradiceardia	<i>C. difficile</i> meningitis bradycardia

Table 1. Illustrative watermark-induced terminology errors drawn from our judge audits and paired qualitative reviews of watermarked outputs. Matched unwatermarked completions on the same questions use the correct terms.

Correct answers hide broken reasoning. The *faulty-but-correct* rate (F&C, correct letter with ≥ 1 flagged defect) climbs from 11.4% to 26.3% on PHI-4-14B under SynthID while accuracy drops only -3.2 pp; OPENBIO-LLM-70B F&C similarly more than doubles (12.9% $\rightarrow$ 27.5%, *). F&C, however, only flags inconsistency—not what is wrong.

Watermarking silently corrupts medical language. Even when accuracy and F&C barely move, watermarks inject *fabricated entities*, *misapplied real terms*, and *corrupted spellings* that standard metrics miss (Table 1). On MEDGEMMA-4B/SynthID, Δ F&C is only $+0.4$, yet the model fabricates $+12.6$ entities per 100 questions and misapplies $+24.3$ real terms; on LINGSHU-7B/SynthID, fabrications reach $+39.2/100$. These errors are especially dangerous: fluent, confident text that no spell-checker or final-letter check would flag.

Each scheme damages a different axis. Watermarking is not generic noise (Table 6): PPL drives surface corruption; SynthID and KGW inflate fabrications and term misuse, with vignette-fabrication ratios up to $18\times$ (PHI-4-14B/PPL) and $7.6\times$ (PHI-4-14B/SynthID); AAR produces broad mild effects; DipMark causes the least damage, consistent with its distortion-free guarantee [36], yet at a fixed key still drives a significant F&C increase on PHI-4-14B—the guarantee only holds in expectation over the key and without hash collisions.

Watermarks corrupt visual reporting. The damage is not confined to text: on MedXpertQA-MM, watermarking changes what models claim to see. Across most medical-VLM configurations, image-SUPPORTED visual claims fall while image-CONTRADICTED claims rise *simultaneously* (Figure 3), so the net visual signal (SUPPORTED — CONTRADICTED per question) is eroded from both sides at

Model	Scheme	Δ Acc	Fab/100	Misapp/100	Spell/100	VigFab	F&C	Major%
<i>MedQA</i>								
PHI-4-14B	None	81.4	0.3	4.1	0.1	1.0 $\times$	11.4	—
	AAR	-2.5	+1.8	+4.0	+1.5	14.0$\times$	+6.5	1.9
	DipMark	-0.9	+0.1	+2.6	+0.3	0.8$\times$	+3.6	2.8
	KGW	-0.7	+1.5	+3.6	+2.2	3.0$\times$	+9.5	3.6
	PPL	-8.6	+3.8	+4.3	+69.6	18.0$\times$	+9.6	2.3
	SynthID	-3.2	+6.6	+8.0	+10.6	7.6$\times$	+14.9	3.3
OPENBIO-LLM-70B	None	73.6	0.0	4.5	0.6	1.0 $\times$	12.9	—
	AAR	-0.2	+0.9	+1.5	+0.9	0.9$\times$	+4.4	4.4
	DipMark	-1.1	+0.4	+0.8	+0.3	1.1$\times$	+1.5	3.2
	KGW	-17.4	+1.9	+4.4	+5.2	3.5$\times$	+4.2	5.6
	PPL	-7.7	+2.1	+1.8	+51.5	1.9$\times$	+6.7	6.6
	SynthID	-11.4	+13.6	+6.9	+25.9	6.6$\times$	+14.6	8.7
<i>MedXpertQA-MM</i>								
MEDGEMMA-4B	None	24.1	1.9	11.1	1.0	1.0 $\times$	16.5	—
	AAR	-1.0	+6.5	+8.1	+3.4	3.1$\times$	+1.2	38.3
	DipMark	+0.2	+0.7	+2.0	+0.2	1.2$\times$	+1.6	35.0
	KGW	-2.6	+8.9	+16.2	+15.4	3.1$\times$	+0.6	31.7
	PPL	-1.7	+6.0	+5.0	+43.6	2.1$\times$	0.0	34.7
	SynthID	-3.1	+12.6	+24.3	+18.0	4.2$\times$	+0.4	36.0
LINGSHU-7B	None	23.9	1.5	12.1	0.4	1.0 $\times$	13.7	—
	AAR	-0.5	+12.9	+18.1	+5.4	2.8$\times$	+3.7	33.0
	DipMark	+1.0	+0.3	+2.7	+0.1	1.0$\times$	+1.4	22.6
	KGW	-0.8	+13.2	+2.4	+16.2	3.6$\times$	+6.0	42.9
	PPL	-0.6	+3.9	+11.5	+37.9	2.4$\times$	+2.3	20.0
	SynthID	-2.1	+39.2	+20.2	+18.9	7.0$\times$	+5.5	30.5

Table 2. Reasoning damage at 99% TPR, for the two most-affected models per benchmark; full results in App. C.1. Columns: accuracy (Δ Acc), fabricated facts (Fab/100), misapplied medical terms (Misapp/100), corrupted spellings (Spell/100), vignette fabrication ratio (VigFab), faulty-but-correct rate (F&C), and major-divergence rate (Major%). Unwatermarked rows (Scheme = None) show absolute values; watermarked rows show Δ /ratio vs. unwatermarked. **Bold:** $p < 0.01$.

once—the model drops grounded observations *and* asserts more claims the image refutes. On MEDGEMMA-4B under KGW this net signal slides from a $+0.10$ baseline through zero to -0.21 at high detectability (TPR $\approx 97\%$, Figure 7): the sign flips, so the watermarked model contradicts the image more often than it supports it. The effect concentrates in medical-specialised and smaller VLMs, while larger general-purpose VLMs (QWEN-3-VL-8/32B) show no significant change (App. C and Figure 10); as on the text axes, accuracy stays within its CI throughout, so a letter-only benchmark registers none of it.

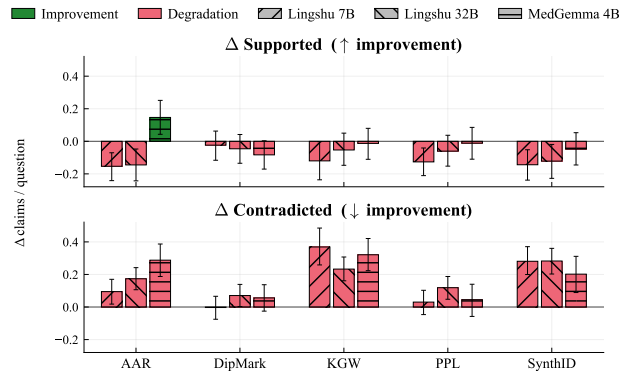


Figure 3. Impact of watermarking on visual understanding. Difference between watermarked and unwatermarked reasoning in average number of supported (top) and contradicted (bottom) visual claims on MedXpertQA.

Specialised and smaller models are most vulnerable; harm grows scheme-specifically. At matched scale and TPR, general-purpose models are less degraded than medical-specialised or small ones: LLAMA-3.1-70B is stable under all five schemes while both same-size medical 70Bs degrade on at least one (similar split among 30B VLMs, App. C.1). The quality-detectability trajectory is also scheme-specific (App. C and Figure 7)—KGW degrades smoothly, while SynthID matches the baseline up to 90% TPR@1%FPR and then drops. At higher distortion, even LLAMA-3.1-70B’s F&C nearly doubles and GEMMA-4-31B’s vignette fabrication reaches 6×.

4.3. Judge Validation

A board-certified physician annotated 650 blinded completions across four (model, scheme) configurations and CLAUDE-SONNET-4.6 independently scored the same samples; on the four axes underlying our headline claims, judge–physician agreement is substantial and judge–judge agreement matches or exceeds it (App. D).

5. Conclusion and Limitations

We introduce a rigorous and systematic evaluation of watermark-induced degradation in medicine, with three specific LLM-as-judges validated by clinicians. We find that, with most watermarks and models tested, the accuracy on medical benchmarks barely drops even at high watermark strength. Nonetheless, our analysis of reasoning traces tells a different story. Watermarks may severely degrade the reasoning of the models (e.g., hallucinate facts, misinterpret data) and may even harm their visual understanding. For reasoning models in particular, we suggest watermarking only the final answer, which we show prevents such degradation.

Limitations and future work. Our benchmarks are multiple-choice and single-turn, leaving open-ended workflows (report writing, multi-turn dialogue, RAG/tool use) and closed-weight models out of scope; whether degradation compounds or attenuates in such settings is an open question. The schemes we test are also model- and domain-agnostic, so future work should explore domain-adapted watermarks that preserve clinically critical tokens.

Impact Statement

LLM watermarking, increasingly mandated by regulation, can silently degrade clinical reasoning in ways standard benchmarks miss—producing fabricated entities, corrupted terminology, and hallucinated findings in fluent, confident text—posing direct patient-safety risks as LLMs enter clinical workflows. Watermarking schemes should be developed and evaluated against domain-specific requirements, and

regulatory frameworks should adopt risk-stratified policies for safety-critical domains rather than treating all uniformly.

Acknowledgements

Robin Staab has been supported by the SERI grant SAFEAI (Certified Safe, Fair and Robust Artificial Intelligence, contract no. MB22.00088). Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or European Commission. Neither the European Union nor the European Commission can be held responsible for them. The work has received funding from the Swiss State Secretariat for Education, Research and Innovation (SERI) (SERI-funded ERC Consolidator Grant). We thank Dr. Andrea Radebold (Internal Medicine, Quinncipiac Medical of Branford/Yale School of Medicine) for her contribution to data annotation.

References

- [1] Aaronson, S. Watermarking of large language models. In *Workshop on Large Language Models and Transformers*, Simons Institute, UC Berkeley, 2023.
- [2] Abdin, M. et al. Phi-4 technical report. *arXiv preprint arXiv:2412.08905*, 2024.
- [3] American Medical Association. 2026 physician survey on augmented intelligence. <https://www.ama-assn.org/system/files/physician-ai-sentiment-report.pdf>, 2026. Accessed: 2026-04-30.
- [4] Arora, R. K., Wei, J., Hicks, R. S., et al. HealthBench: Evaluating large language models towards improved human health. *arXiv preprint arXiv:2505.08775*, 2025.
- [5] Chen, L., Bian, Y., Deng, Y., Cai, D., Li, S., Zhao, P., and Wong, K.-F. WatME: Towards lossless watermarking through lexical redundancy. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 9166–9180, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.496. URL <https://aclanthology.org/2024.acl-long.496/>.
- [6] Chen, P., Ye, J., Wang, G., Li, Y., Deng, Z., et al. GMAI-MMBench: A comprehensive multimodal evaluation benchmark towards general medical AI. *arXiv preprint arXiv:2408.03361*, 2024.
- [7] Dathathri, S., See, A., Ghaisas, S., Huang, P.-S., McAdam, R., Welbl, J., Bachani, V., Kaskasoli, A., Stanforth, R., Matejovicova, T., et al. Scalable watermarking for identifying large language model outputs. *Nature*, 634(8035):818–823, 2024.
- [8] European Commission. Second draft code of practice on marking and labelling of AI-generated content. <https://digital-strategy.ec.europa.eu/en/library/commission-publishes-second-draft-code-practice-marking-and-labelling-ai-generated-content>, 2026. Published: 2026-03-05. Accessed: 2026-04-30.
- [9] European Union. Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (AI Act). Official Journal of the European Union, L 2024/1689, 2024.

- [10] Gemma Team. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.
- [11] Gloaguen, T., Staab, R., Jovanović, N., and Vechev, M. A unified framework for llm watermarks, 2026. URL <https://arxiv.org/abs/2602.06754>.
- [12] Google Open Source. Gemma 4: Expanding the gemmaverse with Apache 2.0, April 2026. URL <https://opensource.googleblog.com/2026/03/gemma-4-expanding-the-gemma-verse-with-apache-20.html>. Google Open Source Blog.
- [13] Gowda, A. S., Chhabria, M. C., and Lam, R. C. Use of large language models on radiology reports: A scoping review. *Journal of the American College of Radiology*, 23(3):437–454, 2026. doi: 10.1016/j.jacr.2025.10.005. Published online 2025.
- [14] Grattafiori, A. et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- [15] Guo, D., Yang, D., Zhang, H., Song, J., Wang, P., Zhu, Q., Xu, R., Zhang, R., Ma, S., Bi, X., et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *Nature*, 645:633–638, 2025. doi: 10.1038/s41586-025-09422-z.
- [16] Hager, P., Jungmann, F., Holland, R., Bhagat, K., Hubrecht, I., Knauer, M., Vielhauer, J., Makowski, M., Braren, R., Kaissis, G., and Rueckert, D. Evaluation and mitigation of the limitations of large language models in clinical decision-making. *Nature Medicine*, 30:2613–2622, 2024. doi: 10.1038/s41591-024-03097-1.
- [17] Hastuti, R. P., Rajagade, R. A., Al Ghanim, M., Zheng, M., and Lou, Q. Factuality beyond coherence: Evaluating LLM watermarking methods for medical texts. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 15129–15147, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-emnlp.818. arXiv:2509.07755.
- [18] Hu, Y. et al. Omnimedvqa: A new large-scale comprehensive evaluation benchmark for medical llm. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [19] Jin, D., Pan, E., Oufattole, N., Weng, W.-H., Fang, H., and Szolovits, P. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021.
- [20] Kim, S. and Yoon, H.-J. Questioning our questions: How well do medical QA benchmarks evaluate clinical capabilities of language models? In *Proceedings of the 24th Workshop on Biomedical Language Processing (BioNLP)*, pp. 274–296, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.bionlp-1.24.
- [21] Kirchenbauer, J., Geiping, J., Wen, Y., Katz, J., Miers, I., and Goldstein, T. A watermark for large language models. In *International Conference on Machine Learning*, pp. 17061–17084. PMLR, 2023.
- [22] Kocaman, V., Kaya, M. A., Feier, A. M., and Talby, D. Clinical large language model evaluation by expert review (CLEVER): Framework development and validation. *JMIR AI*, 4(1):e72153, 2025. doi: 10.2196/72153.
- [23] Kudritipudi, R., Thickett, J., Hashimoto, T., and Liang, P. Robust distortion-free watermarks for language models. 2024.
- [24] Lee, T., Hong, S., Ahn, J., Hong, I., Lee, H., Yun, S., Shin, J., and Kim, G. Who wrote this code? watermarking for code generation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL), Volume 1: Long Papers*, 2024.
- [25] Liu, A., Pan, L., Lu, Y., Li, J., Hu, X., Wen, L., King, I., and Yu, P. S. A survey of text watermarking in the era of large language models. *ACM Computing Surveys*, 57: 47:1–47:36, 2024. doi: 10.1145/3691626. URL <https://arxiv.org/abs/2312.07913>.
- [26] Liu, S., Li, X., Liu, H., Yan, Y., Duan, B., Zheng, Q., Fang, D., Su, L., and Hu, X. Distilling the thought, watermarking the answer: A principle semantic guided watermark for reasoning large language models. *OpenReview*, 2026. URL <https://openreview.net/forum?id=T6NVogsXCZ>. ICLR 2026 Poster.
- [27] Maharana, U., Verma, S., Agarwal, A., Mruthyunjaya, P., Mahapatra, D., Ahmed, S., and Mandal, M. Right prediction, wrong reasoning: Uncovering LLM misalignment in RA disease diagnosis. *arXiv preprint arXiv:2504.06581*, 2025.
- [28] Moll, J., Graf, M., Lemke, T., Lenhart, N., Truhn, D., Delbrouck, J.-B., Pan, J., Rueckert, D., Adams, L. C., and Bressen, K. K. Evaluating reasoning faithfulness in medical vision-language models using multimodal perturbations. In *Proceedings of Machine Learning for Health (MLAH)*, volume 297 of *Proceedings of Machine Learning Research*, 2025. arXiv:2510.11196v2.
- [29] Pal, A. and Sankarasubbu, M. OpenBioLLMs: Advancing open-source large language models for healthcare and life sciences. <https://huggingface.co/aaditya/Llama3-OpenBioLLM-70B>, 2024.
- [30] Pal, A., Umapathi, L. K., and Sankarasubbu, M. Med-HALT: Medical domain hallucination test for large language models. In *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*, pp. 314–334, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.conll-1.21.
- [31] Qwen Team. Qwen3 technical report. *arXiv preprint*, 2025.
- [32] Qwen Team. Qwen3.5: Towards native multimodal agents, February 2026. URL <https://qwen.ai/blog?id=qwen3.5>. Blog post.
- [33] Sellergren, A., Kazemzadeh, S., Jaroensri, T., Kiraly, A., Traverse, M., Kohlberger, T., Xu, S., Jamil, F., Hughes, C., Lau, C., et al. MedGemma technical report. *arXiv preprint arXiv:2507.05201*, 2025.
- [34] Tu, S., Sun, Y., Bai, Y., Yu, J., Hou, L., and Li, J. WaterBench: Towards holistic evaluation of watermarks for large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand, August 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.acl-long.83/>.
- [35] Weissman, G. E., Mankowitz, T., and Kanter, G. P. Unregulated large language models produce medical device-like output. *npj Digital Medicine*, 8(1):148, 2025. doi: 10.1038/s41746-025-01544-y.
- [36] Wu, Y., Hu, Z., Zhang, H., and Huang, H. Dipmark: A stealthy, efficient and resilient watermark for large language models. 2023.
- [37] Xu, W., Chan, H. P., Li, L., Aljunied, M., Yuan, R., Wang, J., Xiao, C., Chen, G., Liu, C., Li, Z., et al. Lingshu: A generalist foundation model for unified multimodal medical understanding and reasoning. *arXiv preprint arXiv:2506.07044*, 2025.

- [38] Yang, P., Li, X., Ni, W., Yin, J., Wang, H., Nan, G., Wang, S., Huang, Y., and Qi, T. Enhancing watermarking quality for LLMs via contextual generation states awareness. *arXiv preprint arXiv:2506.07403*, 2025.
- [39] Zhang, K., Zeng, S., Hua, E., Ding, N., Chen, Z.-R., Ma, Z., Li, H., Cui, G., Qi, B., Zhu, X., Lv, X., Hu, J.-F., Liu, Z., and Zhou, B. UltraMedical: Building specialized generalists in biomedicine. In *Advances in Neural Information Processing Systems 37: Datasets and Benchmarks Track (NeurIPS)*, 2024. Spotlight; arXiv:2406.03949.
- [40] Zuo, Y., Qu, S., Li, Y., Chen, Z.-R., Zhu, X., Hua, E., Zhang, K., Ding, N., and Zhou, B. MedXpertQA: Benchmarking expert-level medical reasoning and understanding. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 80961–80990. PMLR, 2025. arXiv:2501.18362.

A. Additional Experimental Details

A.1. Benchmarks

MedQA [19] contains 2,000 four-option, multiple-choice questions drawn from the United States medical licensing examinations. A meta-analysis by Kim & Yoon [20] identifies MedQA as the benchmark most predictive of real-world clinical performance among existing medical QA datasets.

MedXpertQA-MM [40] comprises 2,000 expert-curated, five-option questions spanning 17 medical specialties. Each question is paired with one to six images (on average 1.43, across 10 modalities including radiology, pathology, and schematic diagrams), requiring joint interpretation alongside clinical vignettes and structured patient data such as laboratory results.

A.2. Models

We evaluate three groups of models. On *MedQA*, we include six instruction-tuned LLMs: 4 general-purpose models (LLAMA-3.1-8B, LLAMA-3.1-70B [14], GEMMA-3-12B [10], and PHI-4-14B [2]), and 2 medical-specialised models (OPENBIOLLM-70B [29], and ULTRAMEDICAL-70B [39]), as well as 4 reasoning models (DEEPSEEK-R1-DISTILL-LLAMA-8B, DEEPSEEK-R1-DISTILL-LLAMA-70B, DEEPSEEK-R1-DISTILL-QWEN-32B [15], and QWEN-3.5-27B [32]). On *MedXpertQA-MM*, we evaluate 7 VLMs spanning 4B–32B parameters: QWEN-3-VL-8B and QWEN-3-VL-32B [31], GEMMA-4-31B [12], LINGSHU-7B and LINGSHU-32B [37], and MEDGEMMA-4B and MEDGEMMA-27B [33]. In particular, LINGSHU-7B, LINGSHU-32B, MEDGEMMA-4B, and MEDGEMMA-27B are medical-specialised VLMs and were not trained on MedXpertQA.

A.3. Watermarking Schemes Overview

Five prominent generation-time watermarking schemes are evaluated, spanning the major algorithmic families: KGW [21] (soft logit bias), DipMark [36] (CDF reweighting), AAR [1] (Gumbel-max sampling), PPL [11] (perplexity-budgeted argmax), and SynthID [7] (tournament sampling). All five share a common structure: at each token position, a context-dependent hash assigns pseudorandom scores to vocabulary entries, and the sampling distribution is modified accordingly. The schemes fall into two categories. *Distortion-based* methods (KGW, PPL at high strength) bias next-token distributions toward high-scoring tokens, introducing a systematic shift away from the model’s original distribution. *Distortion-free* methods (DipMark, AAR at zero strength, SynthID with binary tournaments) preserve the original distribution in expectation over the random watermark key. For each scheme, we sweep multiple strength levels to evaluate their detectability–quality trade-off. We describe each scheme in-depth in App. A.4.

A.4. Watermarking Scheme Details

All five schemes share a common structure: at each token position, a hash of the preceding context seeds a pseudorandom number generator that assigns a score g_u to every vocabulary entry u . The *sampling mechanism* then maps the original next-token distribution p and the scores g to a watermarked next-token distribution $q(g)$, from which the next token is drawn. The schemes differ in the score distribution, in this mapping, and consequently in their detectability–quality trade-offs. We refer to Gloaguen et al. [11] for a unified optimization-theoretic treatment of several of the schemes below. In all cases the score assignment and the reweighting are restricted to the top- k of p as configured in the vLLM sampling parameters.

KGW [21]. The context-dependent hash partitions the top- k vocabulary into a *green list* and a *red list* via i.i.d. Bernoulli($\frac{1}{2}$) draws ($g_u \in \{0, 1\}$), so that in expectation half of the tokens are green. A constant logit bias δ is added to the green-list tokens before sampling, yielding the soft logit-bias watermark

$$q(g) \propto p \exp(\delta g). \quad (1)$$

This is the soft Kirchenbauer scheme; we do not use the hard variant that excludes red-list tokens entirely. Detection tests whether the proportion of green-list tokens among the generated text exceeds the $\frac{1}{2}$ chance level via a one-sided binomial test (and a z -test for cross-checking). Watermark strength is controlled by δ .

DipMark [36]. A distribution-preserving scheme parameterized by $\alpha \in (0, \frac{1}{2}]$. The context hash seeds a uniform-random permutation π of the top- k vocabulary; tokens are then reweighted according to the symmetric α -shift of the permuted CDF. Concretely, in the permuted ordering the scheme averages two indicator-style reweights at the boundaries α and $1 - \alpha$ of the

cumulative mass, which at $\alpha = \frac{1}{2}$ yields exact distortion-freeness in expectation, $\mathbb{E}_G[q(G)] = p$, and as $\alpha \rightarrow 0$ progressively biases mass toward the upper half of the permutation. Detection compares each generated token’s rank under the recomputed permutation against the threshold γ : tokens above the threshold count as “green”, and we apply a one-sided binomial test (and a z -test) against $1 - \gamma$. We evaluate only the distortion-free configuration $\alpha = \gamma = 0.5$, as is standard in the literature.

AAR [1]. For each token position, i.i.d. Gumbel-distributed scores g_u (location 0, scale $\beta = 1$) are drawn and the next token is selected deterministically as

$$\hat{u} = \arg \max_u (g_u + \tau \log p_u), \quad \tau = \frac{\beta}{1 + \beta \delta}, \quad (2)$$

and represented as a Dirac distribution at $\hat{u}$. We fix $\beta = 1$ throughout and sweep $\delta \geq 0$ as the watermark-strength knob. At $\delta = 0$ we recover the canonical AAR sampler $\arg \max_u (g_u + \log p_u)$: by the Gumbel-max trick the marginal distribution of $\hat{u}$ over the score draw is exactly p , so the scheme is distortion-free, and AAR is the optimal sampling mechanism among all distortion-free watermarks that use a sum-based detector [11], corresponding to the soft KL constraint $\text{KL}(\mathbb{E}_G[q(G)] \| p) = 0$. For $\delta > 0$, $\tau < 1$ down-weights $\log p$ relative to the Gumbel score, biasing the argmax toward the score sequence and yielding a more easily detectable but no-longer-distortion-free signal.

PPL [11]. For each token position, the scheme draws i.i.d. Bernoulli-sum scores $g_u \sim \text{Binomial}(N, \frac{1}{2})$ and emits a Dirac distribution at the *tilted argmax*

$$\hat{u}(\beta) = \arg \max_u (g_u + \beta \log p_u), \quad (3)$$

where the tilt $\beta \geq 0$ is solved per token by 1D bisection so that the expected log-likelihood of $\hat{u}$ exactly saturates the per-token log-perplexity budget,

$$\mathbb{E}_G[\log p_{\hat{u}(\beta)}] = \sum_v p_v \log p_v - \varepsilon. \quad (4)$$

The constraint is the same per-token log-perplexity budget $(p - q(g)) \cdot \log p \leq \varepsilon$ as the *hard* variant from the unified framework, but enforced *in expectation over the scores* via the scalar tilt β rather than by a per-realization linear program; in practice this matches the constraint while admitting the simple Gumbel-style implementation. The expectation in the bisection is approximated by 128 Monte Carlo score draws, with a fixed schedule of 60 bisection iterations and the search bracket $\beta \in [0, 100]$. Watermark strength is controlled by ε (larger ε permits more deviation from the model’s distribution and thus a stronger watermark).

SynthID [7]. For each token, m independent layers of i.i.d. Bernoulli scores $g^{(i)} \in \{0, 1\}^{|V|}$ are drawn (in our configuration $m = 30$). Starting from $q^{(0)} := p$, the algorithm applies m tournament reweighting layers parameterized by the number of leaves per match ℓ , controlling the strength of the watermark. With $\ell = 2$ (*non-distortionary* tournament), each layer applies the multiplicative update

$$q_u^{(i)} = q_u^{(i-1)} (1 + g_u^{(i)} - q^{(i-1)} \cdot g^{(i)}), \quad (5)$$

which preserves the marginal distribution $\mathbb{E}_{g^{(i)}}[q^{(i)}] = q^{(i-1)}$. With $\ell > 2$ leaves per match (*distortionary* tournament), the layer instead applies the closed-form tournament-winner reweight

$$q_u^{(i)} = q_u^{(i-1)} \cdot \begin{cases} \frac{1 - (1 - m_i)^\ell}{m_i} & \text{if } g_u^{(i)} = 1, \\ (1 - m_i)^{\ell-1} & \text{if } g_u^{(i)} = 0, \end{cases} \quad m_i := q^{(i-1)} \cdot g^{(i)}, \quad (6)$$

which biases each layer toward tokens with $g_u^{(i)} = 1$ and is no longer distortion-free for $\ell > 2$. Stacking layers progressively strengthens the watermark signal. The original SynthID-Text scheme of Dathathri et al. [7] is typically configured with $\ell = 2$ leaves (the non-distortionary tournament) and a depth in the range $m \in \{20, 30\}$, with detectability controlled by varying m .

A.5. Generation and Reproducibility

All completions are generated with vLLM (v0.19.1), with the watermark realised as a custom LogitsProcessor so that watermarked and unwatermarked runs differ only at the logit-modification step. All experiments use the model-native

tokenizer without post-processing. Decoding uses temperature 0.7 and unrestricted top- k throughout. Each (model, watermark, strength) configuration is run with a fixed sampling seed (default 0); paired watermarked/unwatermarked completions for a given question therefore share both prompt and seed, isolating the watermark as the only source of divergence. The watermark uses output-only context hashing, so the first 4 generated tokens (one context window) are not watermarked; this affects $<1\%$ of tokens and is consistent between generation and detection.

For non-reasoning models we cap output at 2,500 tokens, while for reasoning models we allocate a generation budget of 10,000 tokens, and post-hoc split out the answer portion (after `</think>`) with a maximum of 2,500 tokens for comparability with the non-reasoning pipeline; the watermark is applied to both reasoning trace and answer tokens by default. In practice, the 2,500-token cap never truncated a well-formed response; the only completions reaching the limit were degenerate outputs (e.g. repetitive token loops), which are captured by the format/termination failure axis of the single-response audit. Answer letters are extracted by a two-tier regex parser. A primary pass scans for structured formats observed across models – ANSWER: X, FINAL ANSWER: X, ANSWER: (X), ANSWER: <X>, \boxed{X}, FINAL ANSWER is X, and FINAL ANSWER: <free text> (X) where exactly one letter appears parenthesised. If, and only if, the primary pass finds nothing, a fallback pass admits conservative natural-language patterns (e.g. “the answer is X”, “I would choose X”, or a candidate name followed by a parenthesised letter at end of line). The parser only commits to a letter when exactly one option (A–E) is unambiguously indicated; otherwise the response is recorded as unanswered.

All experiments run on a single node with up to 8 NVIDIA RTX Blackwell 6000 Pro GPUs (96 GB VRAM each). Watermark implementations, generation and analysis pipelines are released alongside the paper to enable reproduction of the reported numbers.

A.6. Existing Asset Licenses

We use existing datasets and model checkpoints only for evaluation, and cite the original creators in Sec. 3.1. The licenses below are the license names stated by the corresponding official repositories or model cards.

Datasets.

- *MedQA*: MIT License.
- *MedXpertQA-MM*: MIT License.

Models.

- LLAMA-3.1-8B and LLAMA-3.1-70B: Meta Llama 3 Community License Agreement.
- BIO-MISTRAL-7B: Apache License 2.0.
- GEMMA-3-12B: Gemma Terms of Use.
- PHI-4-14B: MIT License.
- OPENBIO-LLM-70B and ULTRAMEDICAL-70B: Meta Llama 3 Community License Agreement.
- DEEPSEEK-R1-DISTILL-LLAMA-8B, DEEPSEEK-R1-DISTILL-LLAMA-70B, and DEEPSEEK-R1-DISTILL-QWEN-32B: MIT License; the Llama-derived variants also inherit the applicable Meta Llama Community License terms, and the Qwen-derived variant inherits the applicable Apache License 2.0 upstream terms.
- QWEN-3.5-27B, QWEN-3-VL-8B, and QWEN-3-VL-32B: Apache License 2.0.
- GEMMA-4-31B: Apache License 2.0.
- LINGSHU-7B and LINGSHU-32B: MIT License.
- MEDGEMMA-4B and MEDGEMMA-27B: Health AI Developer Foundations Terms of Use.

B. Per-Configuration Accuracy and Perplexity

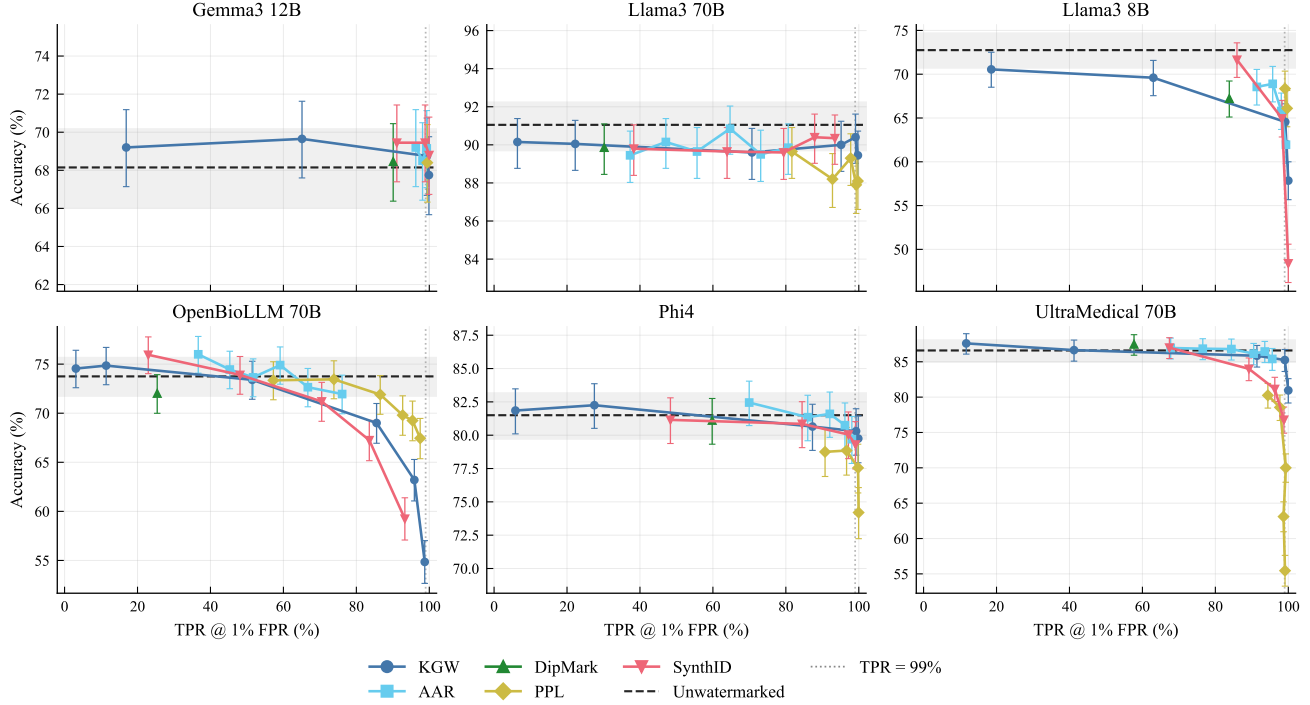


Figure 4. MedQA accuracy vs. detectability (TPR@1% FPR), non-reasoning LLMs across five schemes. Dashed line and grey band: unwatermarked mean and 95% CI; dotted vertical: TPR = 99%. LLAMA-3.1-70B, GEMMA-3-12B, and PHI-4-14B stay within the band across all schemes; the two specialised 70Bs collapse under complementary schemes; LLAMA-3.1-8B degrades broadly.

Tables 3 and 4 report detectability (TPR at 1% FPR) and benchmark accuracy for every (model, scheme, strength) triple evaluated in Sec. 4.1, one benchmark per table. Rows are grouped by scheme and sorted by increasing watermark strength; each model contributes a (TPR, Acc) sub-column pair, with the unwatermarked baseline shown beneath the model header. Configurations strictly stronger than the first to reach $\text{TPR} \geq 99\%$ are omitted (–), as they yield no additional detectability.

B.1. Perplexity Under Watermarking

Figures 5 and 6 plot generation perplexity against watermark detectability. Perplexity increases monotonically with watermark strength on every model and scheme, but its magnitude is a poor predictor of downstream harm: GEMMA-3-12B shows clear perplexity shifts under KGW with no accuracy effect, while ULTRAMEDICAL-70B suffers a 31 pp accuracy collapse under PPL despite only moderate perplexity shifts. Exact per-configuration values accompany the code release.

Table 3. MedQA: watermark detectability (TPR @ 1%FPR, in %) and benchmark accuracy (Acc, in %) for every (model, scheme, strength) configuration evaluated in Sec. 4.1, computed over $N = 2000$ questions (full benchmark). Per-model unwatermarked baseline accuracy is shown beneath each model header. For each (model, scheme) we report all configurations up to and including the first one with TPR $\geq 99\%$; stronger configurations beyond that saturation point are omitted (—), as they yield no additional detectability.

Scheme	Strength	GEMMA-3-12B (68.2)		LLAMA-3.1-70B (91.0)		LLAMA-3.1-8B (72.8)		OPENBioLLM-70B (73.8)		PHI-4-14B (81.5)		ULTRAMEDICAL-70B (86.6)	
		TPR	Acc	TPR	Acc	TPR	Acc	TPR	Acc	TPR	Acc	TPR	Acc
AAR	$\delta = 0$	96.3	69.2	37.3	89.5	91.4	68.5	36.6	76.0	70.0	82.5	67.8	87.0
AAR	$\delta = 0.1$	98.2	68.5	47.1	90.1	95.7	68.9	45.3	74.5	—	—	76.5	86.9
AAR	$\delta = 0.2$	99.3	69.2	55.6	89.6	98.1	65.8	51.7	73.7	86.0	81.3	84.4	86.8
AAR	$\delta = 0.3$	—	—	64.7	90.8	99.4	62.0	59.1	74.9	92.0	81.6	90.5	86.2
AAR	$\delta = 0.4$	—	—	73.1	89.5	—	—	66.7	72.7	96.2	80.8	93.5	86.5
AAR	$\delta = 0.5$	—	—	80.7	89.8	—	—	76.1	72.0	98.1	79.7	95.6	85.4
DipMark	$\alpha = 0.5$	90.1	68.5	30.2	89.8	83.8	67.2	25.4	72.0	59.8	81.1	57.7	87.5
KGW	$\delta = 0.5$	16.9	69.2	6.4	90.1	18.6	70.5	3.1	74.6	5.9	81.8	11.7	87.6
KGW	$\delta = 1$	65.1	69.7	22.2	90.0	63.0	69.6	11.4	74.9	27.5	82.2	41.2	86.7
KGW	$\delta = 2$	99.4	68.8	70.7	89.6	99.2	64.5	51.4	73.4	87.3	80.7	91.4	85.9
KGW	$\delta = 3$	—	—	95.2	90.0	—	—	85.5	69.0	99.2	80.3	99.1	85.2
KGW	$\delta = 4$	—	—	99.1	90.4	—	—	95.9	63.2	—	—	—	—
KGW	$\delta = 5$	—	—	—	—	—	—	98.8	54.9	—	—	—	—
PPL	$\varepsilon = 0$	99.5	68.4	81.7	89.6	99.1	68.3	57.2	73.4	90.8	78.8	94.4	80.2
PPL	$\varepsilon = 0.1$	—	—	92.8	88.2	—	—	73.9	73.5	96.7	78.8	97.7	78.5
PPL	$\varepsilon = 0.2$	—	—	97.8	89.3	—	—	86.6	71.9	—	—	99.3	70.0
PPL	$\varepsilon = 0.3$	—	—	99.3	87.9	—	—	92.7	69.8	99.8	77.5	—	—
PPL	$\varepsilon = 0.4$	—	—	—	—	—	—	95.4	69.2	—	—	—	—
PPL	$\varepsilon = 0.5$	—	—	—	—	—	—	97.5	67.5	—	—	—	—
SynthID	$\ell = 2$	91.1	69.5	38.3	89.8	85.9	71.7	22.9	75.9	48.4	81.2	67.4	87.0
SynthID	$\ell = 3$	98.9	69.5	63.9	89.6	98.3	65.0	48.1	73.9	84.5	80.8	89.2	84.0
SynthID	$\ell = 4$	100.0	68.8	79.4	89.6	100.0	48.4	70.5	71.2	97.2	80.0	96.3	81.2
SynthID	$\ell = 5$	—	—	87.9	90.4	—	—	83.5	67.2	99.1	79.3	98.8	76.8
SynthID	$\ell = 6$	—	—	93.4	90.3	—	—	93.3	59.2	—	—	99.2	63.1

Table 4. MedXpertQA-MM: watermark detectability (TPR @ 1%FPR, in %) and benchmark accuracy (Acc, in %) for every (model, scheme, strength) configuration evaluated in Sec. 4.1, computed over $N = 2000$ questions (full benchmark). Per-model unwatermarked baseline accuracy is shown beneath each model header. For each (model, scheme) we report all configurations up to and including the first one with $\text{TPR} \geq 99\%$; stronger configurations beyond that saturation point are omitted (—), as they yield no additional detectability.

Scheme	Strength	GEMMA-4-31B (54.9)			LINGSHU-32B (29.9)			LINGSHU-7B (24.3)			MEDGEMMA-27B (32.8)			MEDGEMMA-4B (24.4)			QWEN-3-VL-32B (38.9)			QWEN-3-VL-8B (29.8)		
		TPR	Acc	TPR	Acc	TPR	Acc	TPR	Acc	TPR	Acc	TPR	Acc	TPR	Acc	TPR	Acc	TPR	Acc			
AAR	$\delta = 0$	65.2	55.4	72.6	31.4	73.7	23.8	90.0	33.6	84.2	24.2	93.5	39.1	97.9	26.5	—	—	—	—			
AAR	$\delta = 0.1$	71.2	55.0	82.5	29.6	82.9	23.5	94.2	34.2	91.2	24.7	95.5	39.3	98.6	27.0	—	—	—	—			
AAR	$\delta = 0.2$	79.5	54.6	89.5	28.8	91.4	23.4	96.6	33.0	94.0	24.8	97.9	40.4	99.1	27.3	—	—	—	—			
AAR	$\delta = 0.3$	84.8	54.7	93.8	30.1	94.5	24.2	98.0	33.5	96.8	24.0	98.9	39.0	—	—	—	—	—	—			
AAR	$\delta = 0.4$	90.5	54.1	96.2	30.4	97.5	23.8	98.9	32.1	98.2	23.9	99.5	38.7	—	—	—	—	—	—			
AAR	$\delta = 0.5$	92.8	53.6	98.3	30.2	98.5	23.5	99.6	31.9	99.0	22.2	—	—	—	—	—	—	—	—			
DipMark	$\alpha = 0.5$	45.6	54.4	57.0	31.2	57.5	25.7	82.0	32.4	72.5	23.6	87.1	39.0	94.5	29.0	—	—	—	—			
KGW	$\delta = 0.5$	7.1	54.3	9.2	30.6	10.1	24.9	14.8	34.1	14.1	22.1	17.1	38.8	30.3	27.8	—	—	—	—			
KGW	$\delta = 1$	26.9	54.0	34.1	31.8	38.9	23.4	59.2	32.5	54.0	23.2	62.3	40.1	81.2	29.1	—	—	—	—			
KGW	$\delta = 2$	81.0	54.4	91.0	29.5	91.0	22.4	97.0	33.6	91.8	23.8	98.0	39.7	99.2	29.2	—	—	—	—			
KGW	$\delta = 3$	96.9	54.4	99.2	30.4	99.0	25.2	100.0	32.5	96.5	21.5	99.9	38.5	—	—	—	—	—	—			
KGW	$\delta = 4$	99.9	54.0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—			
KGW	$\delta = 5$	—	—	—	—	—	—	—	—	97.2	21.8	—	—	—	—	—	—	—	—			
PPL	$\varepsilon = 0$	88.9	53.3	92.8	30.6	94.7	23.9	98.6	33.1	91.1	22.4	99.1	37.4	99.6	26.9	—	—	—	—			
PPL	$\varepsilon = 0.1$	96.1	52.6	97.8	29.6	97.4	23.2	99.7	32.5	—	—	—	—	—	—	—	—	—	—			
PPL	$\varepsilon = 0.2$	98.8	53.8	99.4	29.2	99.2	23.9	—	—	94.0	21.9	—	—	—	—	—	—	—	—			
PPL	$\varepsilon = 0.3$	99.6	53.4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—			
PPL	$\varepsilon = 0.4$	—	—	—	—	—	—	—	—	96.4	21.1	—	—	—	—	—	—	—	—			
PPL	$\varepsilon = 0.5$	—	—	—	—	—	—	—	—	96.7	21.1	—	—	—	—	—	—	—	—			
SynthID	$\ell = 2$	61.2	53.8	63.3	30.4	69.0	23.8	92.9	31.9	88.6	24.5	89.0	40.0	94.8	26.7	—	—	—	—			
SynthID	$\ell = 3$	81.2	53.8	89.6	30.3	92.5	24.1	98.7	32.9	95.0	22.6	98.1	38.6	99.4	26.4	—	—	—	—			
SynthID	$\ell = 4$	90.7	53.8	98.0	27.9	98.0	24.3	99.9	32.6	96.5	21.6	99.5	39.0	—	—	—	—	—	—			
SynthID	$\ell = 5$	95.4	52.9	99.4	26.6	99.8	23.0	—	—	96.6	20.1	—	—	—	—	—	—	—	—			

Model	Condition	Acc. (%)	TPR (%)	Mean output length (chars)
DEEPSEEK-R1-DISTILL-LLAMA-8B	No WM	59.2	–	9,023
	FA-only, $\delta=1$	61.0	14.3	8,890
	FA-only, $\delta=3$	59.2	87.1	8,856
	FA-only, $\delta=5$	58.9	96.8	8,870
	Full WM, $\delta=1$	61.2	15.4	9,165
	Full WM, $\delta=3$	59.1	93.0	11,621
	Full WM, $\delta=5$	57.2	98.9	15,257
DEEPSEEK-R1-DISTILL-QWEN-32B	No WM	83.6	–	6,447
	FA-only, $\delta=1$	83.6	8.1	6,362
	FA-only, $\delta=3$	82.9	74.3	6,444
	FA-only, $\delta=5$	81.5	97.7	6,482
	Full WM, $\delta=1$	84.1	10.3	6,737
	Full WM, $\delta=3$	83.7	84.4	8,166
	Full WM, $\delta=5$	82.1	99.6	10,027
DEEPSEEK-R1-DISTILL-LLAMA-70B	No WM	92.3	–	6,025
	FA-only, $\delta=1$	92.4	7.0	5,995
	FA-only, $\delta=3$	91.9	59.2	6,072
	FA-only, $\delta=5$	90.8	92.4	6,088
	Full WM, $\delta=1$	91.9	7.1	6,239
	Full WM, $\delta=3$	91.4	82.3	7,227
	Full WM, $\delta=5$	89.8	99.7	8,759
QWEN-3.5-27B	No WM	94.6	–	13,258
	FA-only, $\delta=1$	94.1	23.5	13,125
	FA-only, $\delta=3$	94.5	97.5	13,285
	FA-only, $\delta=5$	94.1	98.8	13,743
	Full WM, $\delta=1$	94.8	26.1	13,285
	Full WM, $\delta=3$	94.0	99.0	14,976
	Full WM, $\delta=5$	77.1	99.5	18,019

Table 5. MedQA accuracy, TPR at 1% FPR, and mean output length (reasoning trace plus final answer, characters) for four reasoning models under No WM, FA-only (post-<thinking> tokens only), and Full WM (entire output), KGW $\delta \in \{1, 3, 5\}$. TPR is computed on the final-answer tokens only.

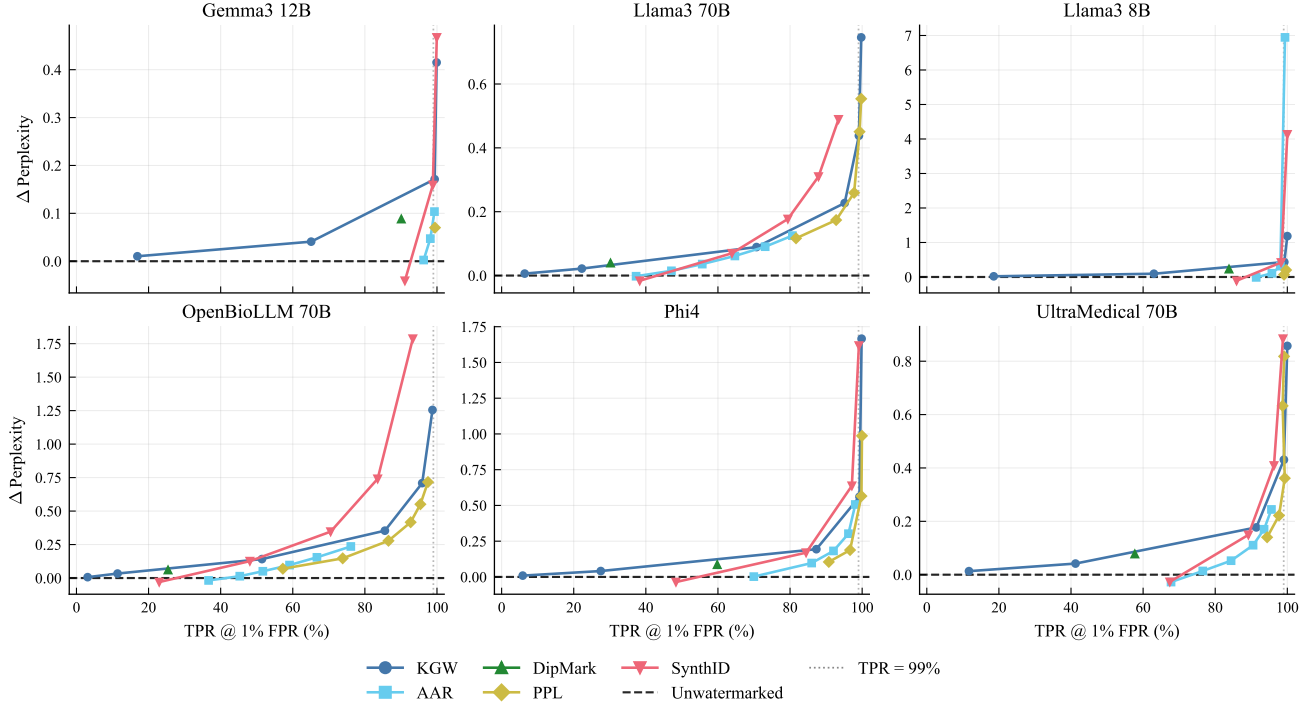


Figure 5. MedQA: generation-perplexity shift vs. watermark detectability, six non-reasoning LLMs. $\Delta \text{PPL} = \text{PPL}_{\text{wm}} - \text{PPL}_{\text{base}}$. Monotonic in strength on every model $\times$ scheme cell; absolute magnitude spans an order of magnitude (note differing y -axes). LLAMA-3.1-8B has the steepest increases; GEMMA-3-12B and ULTRAMEDICAL-70B show contained shifts despite their sharply different accuracy robustness.

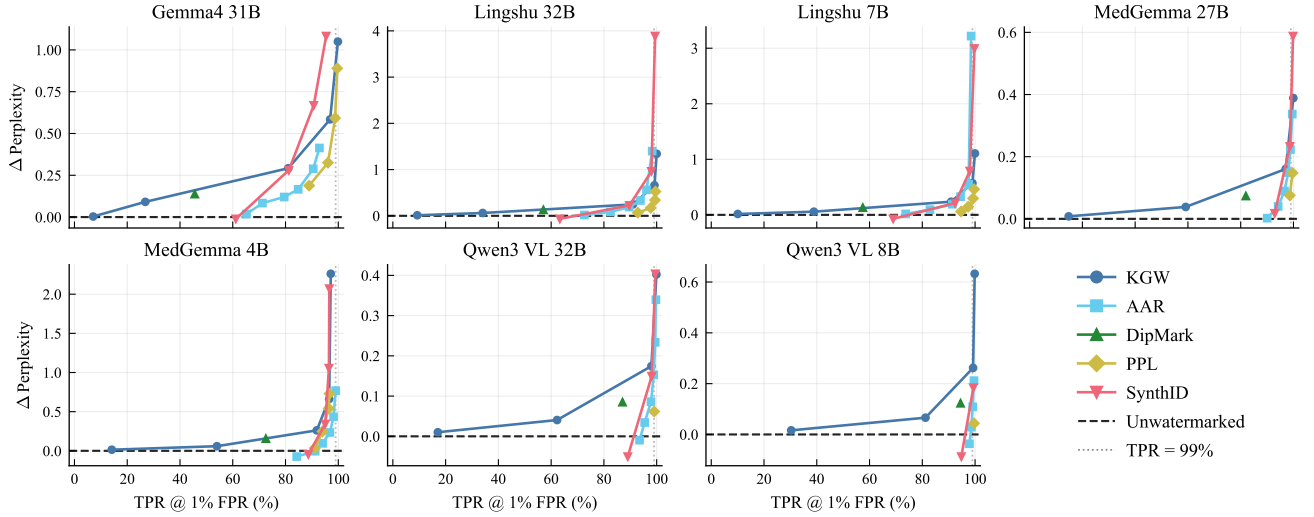


Figure 6. MedXpertQA-MM: generation-perplexity shift vs. watermark detectability, seven VLMs. Same overall shape as MedQA; KGW and SynthID drive the steepest high-TPR increases. LINGSHU-32B and LINGSHU-7B show pronounced jumps at the highest strengths despite minimal accuracy changes (cf. Figure 2).

C. Audit Outcomes Across Watermark Detectability

Sec. 4.2 fixes a single deployment operating point per cell ($\geq 99\%$ TPR), leaving the *shape* of the strength–harm relationship unresolved – whether harm grows smoothly with detectability, or sits near baseline up to a threshold and then jumps. We sweep watermark strength on two contrasting cells: PHI-4-14B on MedQA (an accuracy-stable text-only LLM) and

MEDGEMMA-4B on MedXpertQA-MM (a smaller medical VLM), under the two schemes whose $\geq 99\%$ -TPR signatures differ most in Table 2 – *KGW* (broad multi-axis effect) and *SynthID* (the largest silent-error cell on both models).

Figure 7 plots one harm metric per axis-of-interest against detectability. For PHI-4-14B we report the per-response confident-error rate (a major component of F&C) and the pairwise vignette-fabrication rate (the silent-error signature picked up by VigFab). For MEDGEMMA-4B we report the net visual signal used in Figure 3 and again vignette-fabrication, so the silent-error axis is comparable across the two cells.

KGW vs. SynthID: monotonic slider vs. threshold cliff. KGW degrades smoothly with detectability. On PHI-4-14B, watermarked confident-error climbs from 14.2% at TPR 5.9% to 21.6% at TPR 99.9%, against a stable $\approx 12\%$ unwatermarked baseline; on MEDGEMMA-4B, the net visual signal slides from +0.09 at TPR 14% through zero to -0.21 at TPR 97% (baseline $\approx +0.10$). SynthID is threshold-y: it tracks baseline up to TPR $\approx 90\%$ and then jumps in the final detectability decade. On PHI-4-14B, confident-error is flat at $\approx 13.7\%$ across TPRs 48–84%, then rises to 19.1% (TPR 97%) and 25.0% (TPR 99%); pairwise vignette-fabrication stays $\leq 0.8\%$ up to TPR 97% and jumps to 3.8% at TPR 99%. On MEDGEMMA-4B, vignette-fabrication tracks $5.8\% \rightarrow 7.9\% \rightarrow 18.7\%$ at TPRs 89, 95, 97%.

These shapes have different operational consequences: a KGW deployment can move a slider to trade detectability against quality, whereas a SynthID deployment at the detectability a regulator would require sits past the elbow of a cliff.

C.1. Full Reasoning-Audit Grids

Sec. 4.2 reports only the two most-affected models per benchmark. Below: the full 7-model grids per benchmark, per-scheme FabRatio breakdowns, and the all-VLM companion to Figure 3.

Table 6. **Full reasoning audit at 99% TPR on MedQA**, for all 7 LLMs. We report the accuracy (ΔAcc), the confident-error rate ($\Delta\text{ConfErr}$), the faulty-but-correct rate ($\Delta\text{F&C}$), the percentage of corrupted spellings ($\Delta\text{Spell}/100$), the percentage of fabricated facts ($\Delta\text{Fab}/100$), the within-response contradiction rate (ΔContr), vignette fabrication (ΔVigFab), clue misreads ($\Delta\text{Misread}$), formatting failures (ΔFmt), linguistic corruption (ΔLing), the percentage of misapplied medical terms ($\Delta\text{Misapp}/100$), the percentage of paired responses with major diagnostic divergence (Major%), and the vignette fabrication ratio (FabRatio). For unwatermarked rows we report the actual value, whereas for watermarked rows we report the difference/ratio with the unwatermarked baseline. We **bold** entries that are statistically significant (p-value below 0.01).

Model	Scheme	ΔAcc	$\Delta\text{ConfErr}$	$\Delta\text{F&C}$	$\Delta\text{Spell}/100$	$\Delta\text{Fab}/100$	ΔContr	ΔVigFab	$\Delta\text{Misread}$	ΔFmt	ΔLing	$\Delta\text{Misapp}/100$	Major%	FabRatio
LLAMA-3.1-8B	Unwatermarked	73.3	35.3	29.6	0.8	1.4	22.6	5.8	6.6	0.9	0.1	14.7	—	1.0×
	AAR	-0.6	+8.1	+7.4	+5.8	+4.1	+6.1	+5.5	+3.2	+1.5	+6.1	+0.2	31.9	2.1×
	DipMark	-1.6	-0.7	+1.0	+0.9	+1.0	+3.1	+1.3	+3.4	-0.3	+0.5	-1.5	29.2	1.6×
	KGW	-2.3	+3.0	+2.4	+1.8	+2.2	+5.1	+1.7	+1.6	+0.5	+0.3	-0.3	29.0	1.0×
LLAMA-3.1-70B	PPL	-0.7	+5.4	+4.3	+24.6	+1.0	+4.2	+0.5	+3.6	+0.3	+3.4	+0.7	29.8	1.6×
	SynthID	-2.0	+3.0	+2.2	+2.1	+1.5	+4.0	+3.7	+2.6	+1.5	+0.7	+0.2	33.5	1.2×
	Unwatermarked	91.8	6.9	8.1	0.4	0.1	5.8	0.4	1.0	0.0	0.0	1.5	—	1.0×
	AAR	-1.2	-0.5	+0.1	+0.3	0.0	+0.1	+0.1	+0.5	0.0	+0.1	-0.3	4.1	—
GEMMA-3-12B	DipMark	-1.2	+1.4	+1.5	-0.1	-0.1	+1.4	-0.2	+0.2	+0.2	0.0	0.0	4.0	—
	KGW	-1.3	+1.8	+1.9	+1.0	0.0	+1.6	+0.4	+0.3	+0.1	+0.1	+0.6	5.1	1.0×
	PPL	-1.8	+3.3	+7.5	+89.4	+1.9	+0.9	+0.3	+3.5	+0.6	+30.9	+0.3	5.0	—
	SynthID	-0.9	+2.0	+1.2	+6.5	+0.2	+0.9	+0.4	-0.1	+0.1	+3.4	0.0	5.9	0.0×
OPENBioLLM-70B	Unwatermarked	67.0	29.2	22.5	0.7	1.3	20.4	3.4	4.5	0.0	0.0	9.0	—	1.0×
	AAR	+1.3	+2.5	+3.4	+0.2	+0.4	+3.2	-0.7	+0.3	+0.2	0.0	+3.6	21.0	0.9×
	DipMark	+1.8	-1.0	+1.1	-0.3	+0.2	+1.7	-0.4	-0.7	+0.4	+0.1	+1.5	20.9	0.4×
	KGW	+1.5	+0.7	+1.7	+0.1	0.0	+1.9	+1.0	+0.1	+0.1	0.0	+0.8	22.4	1.3×
OPENBioLLM-70B	PPL	+1.5	+2.5	+3.2	+20.8	+1.2	+2.1	+0.1	+3.9	+0.6	+3.7	+3.4	23.1	1.1×
	SynthID	+1.9	+3.0	+4.5	+2.6	+0.3	+3.0	+1.1	+0.5	+0.2	+0.1	+1.5	22.9	1.1×
	Unwatermarked	73.6	13.9	12.9	0.6	0.0	8.6	2.2	2.2	14.8	0.1	4.5	—	1.0×
	AAR	-0.2	+6.6	+4.4	+0.9	+0.9	+3.8	+1.7	+0.6	+4.0	+0.9	+1.5	20.6	0.9×
Ultramedical 70B	DipMark	-1.1	+2.3	+1.5	+0.3	+0.4	+1.4	+0.8	+1.2	+5.3	+0.1	+0.8	18.6	1.1×
	KGW	-17.4	+11.4	+4.2	+5.2	+1.9	+9.3	+7.0	+6.6	+20.2	+0.3	+4.4	24.6	3.5×
	PPL	-7.7	+7.1	+6.7	+51.5	+2.1	+4.1	+3.9	+4.6	+13.0	+8.5	+1.8	23.0	1.9×
	SynthID	-11.4	+24.0	+14.6	+25.9	+13.6	+12.3	+13.0	+7.9	+15.5	+13.8	+6.9	27.8	6.6×
Phi-4-14B	Unwatermarked	86.6	11.0	11.8	1.0	0.0	8.0	0.5	1.5	23.2	2.2	2.4	—	1.0×
	AAR	-1.5	+2.3	+2.5	+0.5	+0.4	+4.5	+0.9	+1.8	-13.1	-0.9	+0.6	10.7	1.0×
	DipMark	+1.2	+2.0	+1.7	0.0	0.0	+0.9	+0.9	+0.2	-1.7	+0.7	+0.1	10.2	0.5×
	KGW	-1.5	+2.0	+1.8	+0.4	+0.2	+0.9	+1.0	+0.4	+10.3	+1.4	+0.6	10.3	3.0×
Phi-4-14B	PPL	-39.5	+6.5	+0.2	+97.8	+1.3	+2.7	+2.4	+7.4	+27.6	+22.3	+2.6	11.4	22.0×
	SynthID	-9.2	+3.9	+3.3	+6.1	+0.5	+3.5	+1.3	+2.7	+4.1	+4.1	+0.8	10.1	1.2×
	Unwatermarked	81.4	12.3	11.4	0.1	0.3	7.8	1.6	2.0	0.4	0.0	4.1	—	1.0×
	AAR	-2.5	+6.2	+6.5	+1.5	+1.8	+4.4	+2.9	+2.2	+0.6	+0.2	+4.0	11.3	14.0×
BioMISTRAL-7B	DipMark	-0.9	+3.9	+3.6	+0.3	+0.1	+2.0	+0.6	-0.1	+0.2	0.0	+2.6	10.8	0.8×
	KGW	-0.7	+9.2	+9.5	+2.2	+1.5	+5.5	+3.4	+3.6	+0.2	+0.1	+3.6	14.2	3.0×
	PPL	-8.6	+10.0	+9.6	+69.6	+3.8	+5.2	+3.2	+4.0	+5.4	+10.4	+4.3	13.4	18.0×
	SynthID	-3.2	+13.7	+14.9	+10.6	+6.6	+9.4	+6.8	+5.5	+0.7	+3.9	+8.0	15.4	7.6×
BioMISTRAL-7B	Unwatermarked	40.8	1.1	1.3	0.3	0.4	1.8	1.9	0.7	7.4	0.0	0.4	—	1.0×
	AAR	-1.1	+21.2	+8.4	+5.2	+12.3	+8.6	+13.3	+8.2	+7.3	+3.4	+9.1	25.1	14.0×
	DipMark	-1.3	+7.9	+3.4	+0.9	+1.4	+5.8	+3.8	+2.6	-6.5	+0.2	+0.9	11.5	4.9×
	KGW	-38.7	+27.8	-0.6	+19.6	+9.1	+16.9	+13.8	+11.7	+28.3	+3.9	+16.7	48.9	18.1×
BioMISTRAL-7B	PPL	-5.0	+13.2	+5.0	+18.5	+1.9	+8.6	+5.2	+5.3	+11.1	+2.0	+5.6	11.8	9.5×
	SynthID	-30.4	+10.5	+3.4	+82.3	+153.4	+1.8	+30.4	+8.2	+66.7	+69.0	+1.3	63.0	93.0×

Single-response axes (per 100 questions where suffixed): ConfErr (confident error, on BOTH-CORRECT pairs), F&C (faulty-but-correct: correct letter + ≥ 1 defect), Spell/100 (fabricated entities), Contr (within-response contradiction), VigFab (vignette fabrication), Misread (clue misread), Fmt (format/termination failure), Ling (linguistic corruption), Misapp/100 (misapplied real terms). Pairwise: Major% (MAJOR divergence on BOTH-CORRECT pairs), FabRatio (WM/Base vignette-fabrication rate).

Table 7. **Full reasoning audit at 99% TPR on MedXpertQA-MM**, for all 7 VLMs. We report the accuracy (ΔAcc), the confident-error rate ($\Delta\text{ConfErr}$), the faulty-but-correct rate ($\Delta\text{F&C}$), the percentage of corrupted spellings ($\Delta\text{Spell}/100$), the percentage of fabricated facts ($\Delta\text{Fab}/100$), the image-grounding net change ($\Delta\text{Net\%}$), the within-response contradiction rate (ΔContr), vignette fabrication (ΔVigFab), clue misreads ($\Delta\text{Misread}$), formatting failures (ΔFmt), linguistic corruption (ΔLing), the percentage of misapplied medical terms ($\Delta\text{Misapp}/100$), supported visual claims (ΔSup), contradicted visual claims (ΔCon), the percentage of paired responses with major diagnostic divergence (Major%), and the vignette fabrication ratio (FabRatio). For unwatermarked rows we report the actual value, whereas for watermarked rows we report the difference/ratio with the unwatermarked baseline. We **bold** entries that are statistically significant (p-value below 0.01).

Model	Scheme	ΔAcc	$\Delta\text{ConfErr}$	$\Delta\text{F&C}$	$\Delta\text{Spell}/100$	$\Delta\text{Fab}/100$	$\Delta\text{Net\%}$	ΔContr	ΔVigFab	$\Delta\text{Misread}$	ΔFmt	ΔLing	$\Delta\text{Misapp}/100$	ΔSup	ΔCon	Major%	FabRatio
GEMMA-4-31B	Unwatermarked	56.2	12.2	9.2	0.1	0.3	2	5.0	0.8	1.9	0.1	0.0	1.7	2.55	0.53	—	1.0×
	AAR	-0.6	-0.6	-0.3	+0.9	+0.3	+1	+0.7	+0.1	-0.1	+0.1	+0.8	-1.1	-0.01	-0.03	6.0	3.0×
	DipMark	-0.1	-1.0	-0.3	+0.5	+0.2	+2	+0.5	-0.5	0.0	+0.1	0.0	-0.2	+0.02	-0.03	3.6	2.0×
	KGW	+0.2	-1.7	-0.3	+1.8	+0.2	+1	+1.7	-0.2	+0.3	+0.2	+1.5	+0.2	+0.04	+0.01	6.8	0.5×
	PPL	-0.9	+1.6	+3.1	+34.5	+1.5	-2	+1.8	-0.3	+3.4	+2.1	+22.5	0.0	-0.04	-0.01	5.0	6.0×
LINGSU-7B	SynthID	-1.0	+0.4	-0.3	+7.6	+0.8	+3	+1.0	-0.1	-0.3	+0.1	+11.0	-0.9	+0.02	-0.04	6.8	4.0×
	Unwatermarked	23.9	55.3	13.7	0.4	1.5	0	17.0	3.5	7.6	0.8	0.5	12.1	1.15	0.77	—	1.0×
	AAR	-0.5	+13.8	+3.7	+5.4	+12.9	-67	+3.5	+7.9	+3.8	+2.0	+3.5	+18.1	-0.15	+0.09	33.0	2.8×
	DipMark	+1.0	+5.4	+1.4	+0.1	+0.3	-6	+2.7	+0.1	0.0	-0.1	+0.1	+2.7	-0.02	-0.00	22.6	1.0×
	KGW	-0.8	+28.6	+6.0	+16.2	+13.2	-103	+20.2	+10.4	+11.0	+5.4	+4.6	+2.4	-0.12	+0.37	42.9	3.6×
LINGSU-32B	PPL	-0.6	+7.3	+2.3	+37.9	+3.9	-44	+3.9	+3.0	+7.9	+5.1	+8.4	+11.5	-0.13	+0.03	20.0	2.4×
	SynthID	-2.1	+26.2	+5.5	+18.9	+39.2	-105	+13.2	+16.8	+8.5	+7.7	+13.5	+20.2	-0.14	+0.28	30.5	7.0×
	Unwatermarked	31.1	40.8	15.1	0.8	0.1	1	9.6	2.0	3.7	0.4	0.1	5.9	1.38	0.68	—	1.0×
	AAR	-1.8	+9.2	+1.3	+2.9	+5.4	-46	+4.8	+5.5	+3.4	+2.5	+3.5	+2.0	-0.15	+0.17	18.0	3.8×
	DipMark	+0.4	-1.2	-1.0	-0.1	+0.4	-16	+1.0	+0.1	+0.6	-0.2	0.0	0.0	-0.05	+0.07	15.3	0.8×
MEDGEMMA-4B	KGW	-4.0	+10.1	-0.1	+1.9	+1.8	-39	+8.9	+2.4	+4.8	+1.1	+0.8	+3.6	-0.05	+0.23	14.4	1.8×
	PPL	-2.4	-3.2	-2.0	+27.4	+0.8	-26	+1.2	+1.2	+7.1	+6.7	+5.2	-0.6	-0.06	+0.12	11.2	2.2×
	SynthID	-3.8	+6.7	+2.0	+23.9	+13.7	-56	+16.2	+9.8	+9.5	+8.8	+13.2	+5.2	-0.12	+0.28	21.7	4.9×
	Unwatermarked	24.1	56.9	16.5	1.0	1.9	0	29.7	3.5	9.0	7.7	0.0	11.1	1.24	1.11	—	1.0×
	AAR	-1.0	+13.0	+1.2	+3.4	+6.5	-108	+2.1	+7.4	+3.3	+11.2	+4.6	+8.1	-0.15	+0.29	38.3	3.1×
MEDGEMMA-27B	DipMark	+0.2	+11.6	+1.6	+0.2	+0.7	-103	-0.9	+0.1	-0.2	-0.2	+0.3	+2.0	-0.08	+0.06	35.0	1.2×
	KGW	-2.6	+19.2	+0.6	+15.4	+8.9	-266	+11.2	+6.5	+7.4	+13.1	+3.5	+16.2	-0.01	+0.32	31.7	3.1×
	PPL	-1.7	+8.3	0.0	+43.6	+6.0	-46	-0.6	+3.1	+8.2	+8.0	+9.7	+5.0	-0.01	+0.05	34.7	2.1×
	SynthID	-3.1	+24.3	+0.4	+18.0	+12.6	-168	+8.5	+11.1	+8.1	+22.1	+8.7	+24.3	-0.05	+0.20	36.0	4.2×
	Unwatermarked	33.7	41.5	16.9	0.6	2.3	0	12.4	1.6	5.8	2.8	0.0	9.1	1.90	1.59	—	1.0×
QWEN-3-VL-8B	AAR	-0.9	-3.0	-0.8	+1.5	+1.9	-43	+2.6	+2.9	-0.7	+2.6	+0.1	-2.0	-0.08	+0.06	19.5	1.6×
	DipMark	+1.2	-5.8	-0.3	+0.3	-0.2	-16	+0.2	+0.1	-1.1	+0.7	+0.1	-1.0	-0.06	+0.00	18.4	1.0×
	KGW	-1.0	+4.7	+1.9	+0.7	+0.3	-41	+3.3	+1.6	+0.8	+1.6	+0.2	+3.1	-0.01	+0.13	18.5	1.8×
	PPL	-0.2	+7.4	+3.4	+34.2	+2.7	-23	+5.5	+1.5	+4.5	+1.8	+6.7	-3.0	-0.04	+0.04	23.0	1.4×
	SynthID	-0.3	-1.9	-0.8	+1.5	+1.3	+3	+2.7	+0.9	-0.3	+0.3	+0.2	-1.9	+0.01	+0.00	22.8	0.9×
QWEN-3-VL-32B	Unwatermarked	31.3	47.0	19.9	0.6	1.8	0	21.7	3.0	4.9	7.6	0.6	11.0	1.95	1.57	—	1.0×
	AAR	-4.3	0.0	-5.2	0.0	+2.5	-3	+1.3	+0.6	+0.7	+7.3	+0.6	+7.3	+0.06	+0.07	22.2	1.8×
	DipMark	-2.5	+2.9	-1.5	-0.1	+1.8	+17	+1.5	+1.1	+2.6	+1.2	-0.2	+2.3	+0.06	-0.01	25.0	1.1×
	KGW	-1.0	+1.1	+0.7	+0.7	+1.9	+8	+4.5	+2.2	+1.7	+2.9	+0.1	+3.4	+0.07	+0.04	24.6	1.7×
	PPL	-5.1	+2.0	-2.5	+22.5	+1.1	-22	-1.1	+0.6	+4.7	+7.3	+7.2	+4.7	-0.10	-0.02	24.9	1.5×
QWEN-3-VL-72B	SynthID	-4.5	+3.1	-2.1	+0.1	+1.4	-9	+2.0	+1.9	+0.6	+6.2	-0.4	+5.0	-0.04	-0.00	24.5	1.0×
	Unwatermarked	40.8	27.9	15.7	0.2	0.7	1	15.3	1.3	3.8	3.8	0.2	3.3	2.69	1.29	—	1.0×
	AAR	-3.0	+3.1	-0.7	+1.5	+1.9	-2	-1.1	+0.3	-0.1	+0.5	+3.0	+2.7	-0.01	+0.02	15.4	0.6×
	DipMark	-1.4	+5.1	+1.0	+0.5	+0.9	-3	-1.0	+0.1	0.0	+0.6	+0.8	+4.0	-0.00	+0.04	14.1	0.8×
	KGW	-0.9	+3.9	-0.1	+1.1	+0.2	+3	+1.4	+0.8	-0.2	+0.6	+1.2	+2.1	+0.08	+0.04	15.4	1.0×
QWEN-3-VL-128B	PPL	-2.6	+4.1	-0.6	+35.6	+2.0	-3	+2.2	-0.5	+2.6	+2.2	+11.6	+1.6	+0.01	+0.05	14.4	1.2×
	SynthID	+1.1	+3.9	+2.5	+1.2	+2.3	-9	-0.4	+0.9	-0.4	+0.7	+3.5	+3.9	-0.04	+0.08	15.3	1.6×

Single-response axes (per 100 questions where suffixed): ConfErr (confident error, on BOTH-CORRECT pairs), F&C (faulty-but-correct: correct letter + ≥ 1 defect), Spell/100, Fab/100 (fabricated entities), Contr (within-response contradiction), VigFab (vignette fabrication), Misread (clue misread), Fmt (format/termination failure), Ling (linguistic corruption), Misapp/100 (misapplied real terms), Image-grounding (multimodal): Sup, Con (per-question SUPPORTED/CONTRADICTED visual claims), Net% (relative change in SUPPORTED — CONTRADICTED). Pairwise: Major% (MAJOR divergence on BOTH-CORRECT pairs), FabRatio (WM/Base vignette-fabrication rate).

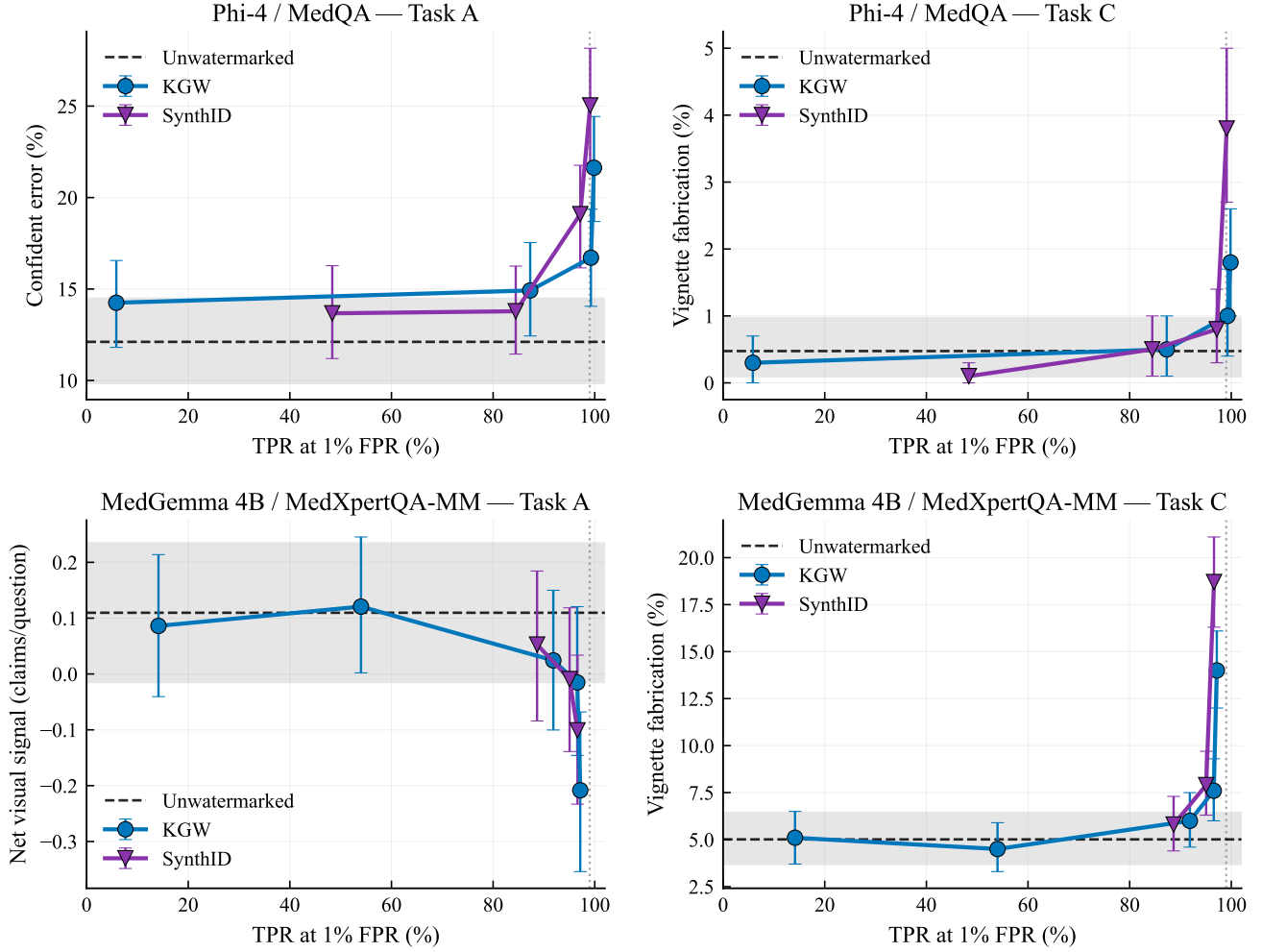


Figure 7. **Audit outcomes vs. watermark detectability** on PHI-4-14B (MedQA, top) and MEDGEMMA-4B (MedXpertQA-MM, bottom) under KGW (blue) and SynthID (purple). Each point is one (scheme, strength); dashed line and grey band give the unwatermarked baseline mean and 95%-bootstrap CI; dotted vertical marks TPR = 99%. Top: per-response confident-error rate restricted to right-letter responses (left); pairwise vignette-fabrication rate (right). Bottom: net visual signal, SUPPORTED — CONTRADICTED visual claims per question (left); pairwise vignette-fabrication rate (right).

C.2. Reasoning Models: Full vs. Answer-Only Watermarking

The two highest-scoring models on MedQA in our setup are reasoning models (QWEN-3.5-27B (94.6%) and DEEPSEEK-R1-DISTILL-LLAMA-70B (92.3%)), narrowly above the strongest instruction-tuned baseline (LLAMA-3-70B (91.0%)). Unlike instruction-tuned models, they generate a reasoning trace (usually hidden) inside `<thinking>` tags before producing the final answer; a watermark can therefore be applied in two distinct ways: *answer-only* (FA-only), which watermarks only the post-`<thinking>` answer tokens, and *full watermark* (Full WM), which watermarks both the reasoning trace and the final answer. We compare both approaches on four reasoning models—DEEPSEEK-R1-DISTILL-LLAMA-8B and DEEPSEEK-R1-DISTILL-LLAMA-70B, DEEPSEEK-R1-DISTILL-QWEN-32B, and QWEN-3.5-27B with KGW at 3 strength levels ($\delta \in \{1, 3, 5\}$).

Full WM degrades reasoning quality on every axis except the final answer. Across the six judge dimensions, Full WM drives substantial negative shifts on *reasoning efficiency* and *terminological precision*, with smaller drops on *clinical coherence* and *distractor engagement* for each of the four models (Figure 11). Two dimensions change negligibly: *answer-reasoning alignment* and *diagnostic accuracy*. The model thus still selects the correct letter through reasoning that is internally consistent with its conclusion, but its output becomes substantially harder to use as a clinical reference due to imprecise or fabricated terminology, redundant deliberation, and distractor engagement that the unwatermarked response

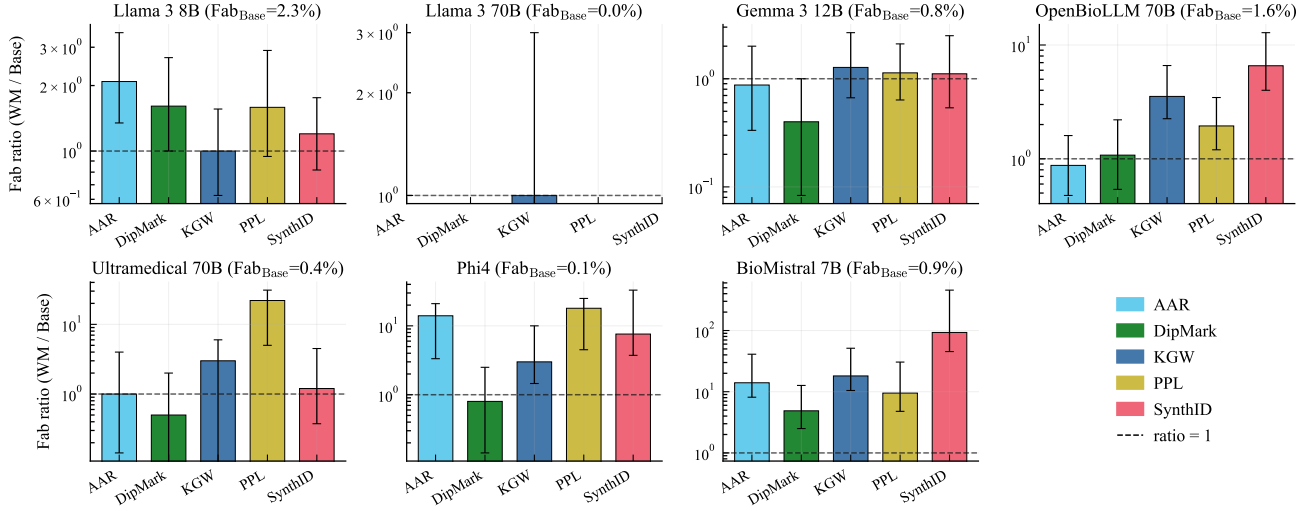


Figure 8. **Pairwise vignette-fabrication ratio per scheme (MedQA, log scale).** WM/Base rate ratio per (model, scheme) cell; dashed line at 1 (no effect). Cells with very low baseline rates have wide CIs.

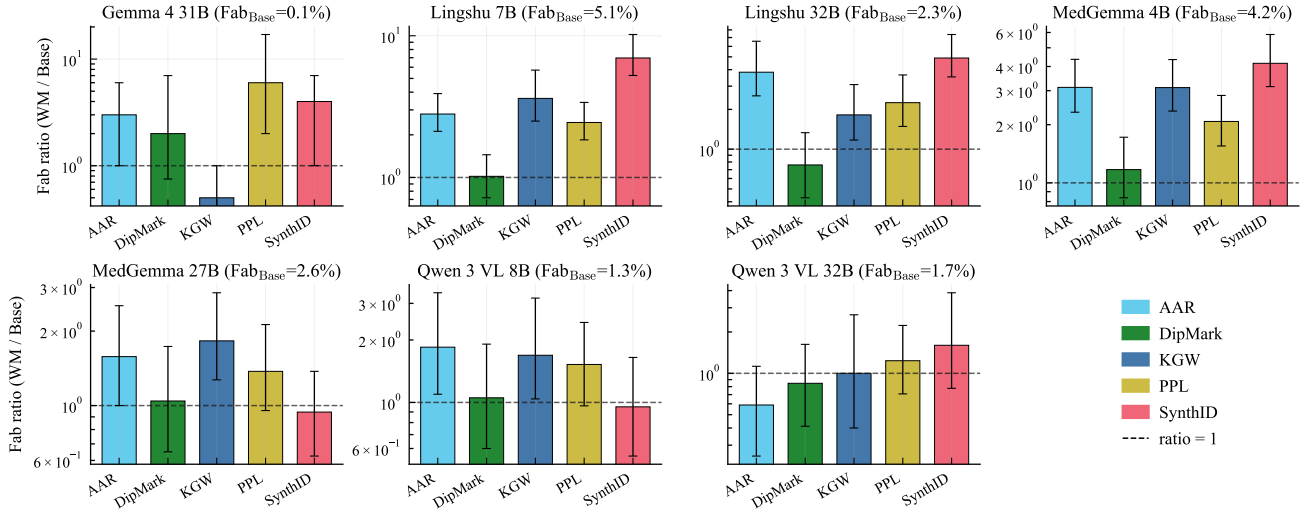


Figure 9. **Pairwise vignette-fabrication ratio per scheme (MedXpertQA-MM, log scale).** Conventions as in Figure 8.

dismisses. We also rate how harmful the reasoning is (NO_HARM_RISK/LOW_RISK/HARM_RISK_PRESENT). Even though both small and large models similarly degrade the reasoning, only small models’ reasoning is judged to be significantly harmful.

Watermarking only the final answer is essentially free. Scoring Full WM against FA-only as the reference reproduces the same per-dimension pattern within bootstrap intervals (Figure 11, orange vs. blue): FA-only is equivalent to Base on every axis. HARM_RISK_PRESENT likewise stays at the No WM baseline under FA-only on every model (Table 9): the watermark, applied only to post-`<thinking>` answer tokens, leaves the model’s chain of thought unperturbed.

Trace watermarking inflates verbosity and induces recursive self-correction loops. Inspection of Full WM traces reveals a consistent pathology across the four models: revisiting already established (and corrected) premises. A representative excerpt is shown in Table 8. Quantitatively, Full WM at $\delta = 5$ inflates mean total output length (reasoning trace plus final answer) by +36 % to +69 % across the four models, while FA-only does not (Table 5). This means that Full WM’s increase in detectability (at a given distortion level) is partly a length artefact rather than a stronger per-token signal.

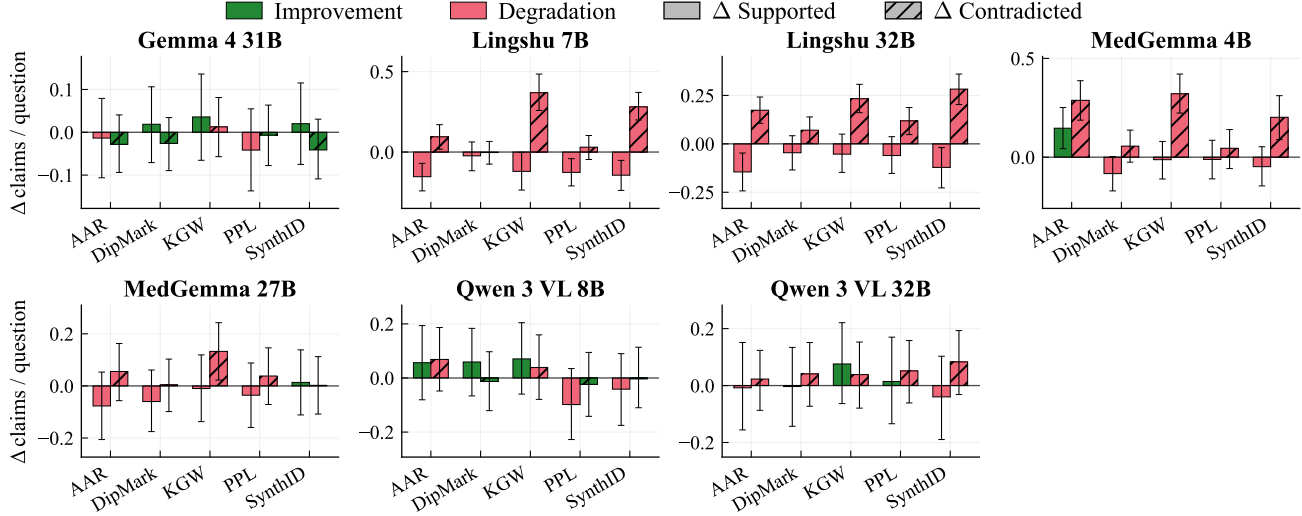


Figure 10. Per-scheme Δ in image-supported and image-contradicted visual claims, all 7 VLMs (MedXpertQA-MM). All-VLM companion to Figure 3. Within each panel the left bar is Δ Supported (solid) and the right bar is Δ Contradicted (hatched). Bars are coloured green where the change improves visual grounding (Δ Supported > 0 or Δ Contradicted < 0) and red otherwise.

Baseline (No WM)

Linear: “The patient has low Vitamin D but normal Ca^{++} , suggesting secondary hyperparathyroidism.”

Full WM (KGW $\delta = 3$)

Circular: “Wait, let me rethink. The Ca^{++} is normal. Actually, the Ca^{++} is not low. Hmm. Let me check the Ca^{++} again.”

Table 8. Excerpt of DEEPSEEK-R1-DISTILL-QWEN-32B reasoning traces, baseline vs. Full WM.

Per-dimension judge scores. Figure 11 shows the mean pairwise Full WM–reference score on all six judge dimensions, with the two reference conditions (vs. Base, vs. FA-only) plotted side by side. The two series overlap within their 95% bootstrap intervals on every panel: this is equivalent to $\text{FA-only} \approx \text{Base}$ on every axis, since $(\text{Full WM} - \text{Base}) - (\text{Full WM} - \text{FA-only}) = \text{FA-only} - \text{Base}$. Across all four models the pattern at $\delta = 5$ is the same: *reasoning efficiency* and *terminological precision* drop substantially, *clinical coherence* and *distractor engagement* drop more mildly, and *answer–reasoning alignment* and *diagnostic accuracy* stay near zero.

Clinical-deployment ratings. Table 9 gives the absolute HARM_RISK_PRESENT and MAJOR_REVISION rates per condition. Only LLAMA-3.1-8B exhibits a Full WM-specific safety jump (+12.8 pp HARM at $\delta = 5$); on the three larger models the Full WM HARM% sits within ± 2 pp of the No WM baseline at every δ .

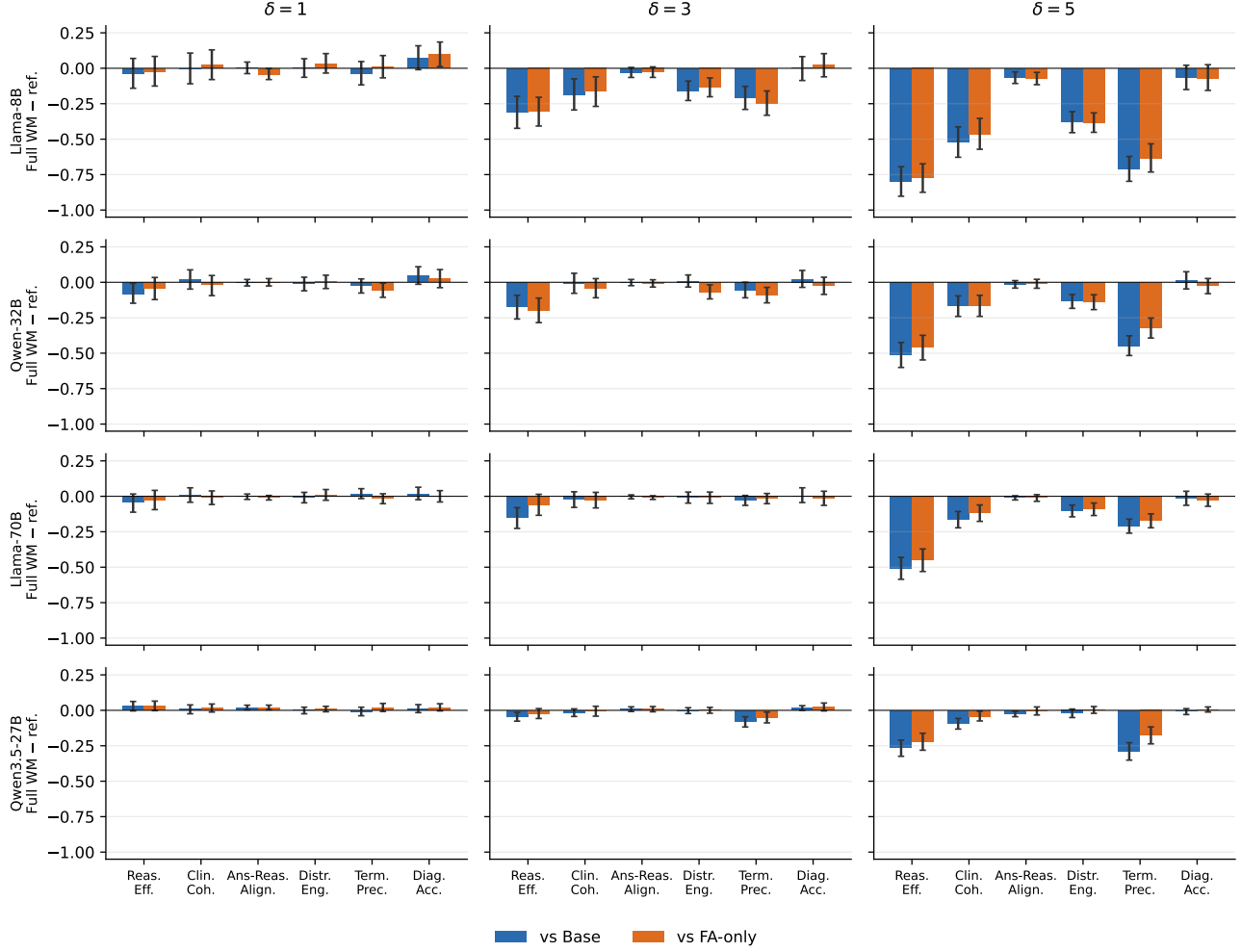


Figure 11. **Pairwise mean Full WM-reference score on six reasoning-quality axes**, four reasoning models $\times$ three KGW strengths. Blue: Full WM scored against the unwatermarked Base response; orange: Full WM scored against the FA-only response. Per-question scores are integers in $\{-2, -1, 0, +1, +2\}$ (rubric in App. E.2.3); panel values are sample means. Negative bar = Full WM judged worse than the reference. Error bars: 95% bootstrap intervals.

Table 9. HARM_RISK_PRESENT and MAJOR_REVISION rates by watermarking scope (KGW). No WM is judged in pairwise mode against Full WM (small contrast bias) and is delta-independent; FA-only and Full WM are absolute per-response ratings.

Model	δ	No WM		FA-only		Full WM	
		HARM%	MAJOR%	HARM%	MAJOR%	HARM%	MAJOR%
LLAMA-8B	1			56.6	56.2	58.7	58.1
	3	50.9	48.8	58.2	57.6	57.1	56.7
	5			55.7	55.3	63.7	63.1
QWEN-32B	1			16.2	16.0	16.2	16.2
	3	22.1	20.7	18.8	18.8	17.5	17.5
	5			18.9	18.6	19.7	19.5
LLAMA-70B	1			8.0	8.0	10.0	10.0
	3	10.0	9.3	8.0	8.0	11.0	11.0
	5			8.8	8.8	9.0	9.0
QWEN-3.5-27B	1			2.8	2.8	2.6	2.6
	3	3.7	3.7	2.4	2.4	1.6	1.6
	5			2.1	2.5	2.6	2.6

D. Judge Validation

A board-certified physician annotated 650 completions blinded to model and watermark configuration, drawn from four (model, scheme) configurations spanning the full error-density range: PHI-4-14B/SynthID and LLAMA-3.1-70B/DipMark on MedQA, LINGSHU-7B/SynthID and GEMMA-4-31B/DipMark on MedXpertQA-MM (250 MedQA + 250 MedXpertQA-MM single-response completions, plus 150 pairwise pairs). As a second automated baseline, CLAUDE-SONNET-4.6 scores the same 650 samples independently.

On the four axes that carry our headline claims, judge–physician agreement is substantial: fabricated-entity counts reach $\kappa=0.75$ / $\rho=0.75$ ($n=500$), misapplied-term counts $\kappa=0.42$ / $\rho=0.47$, image-contradicted claims $\rho=0.74$ ($n=250$, MM only), and pairwise diagnostic divergence $\kappa_w=0.65$ / binary $\kappa=0.67$ ($n=150$). Agreement between GEMINI-3-FLASH and CLAUDE-SONNET-4.6 on the medical-content axes (fabrications, misapplied terms, pairwise divergence) matches or exceeds the corresponding judge–physician values (0.80 / 0.81, 0.50 / 0.52, $\kappa_w=0.66$ / $\kappa=0.72$, respectively), confirming that two independently trained LLM-judges recover the same signal. The full per-axis agreement grid (Table 10) shows that lower κ values concentrate on low-prevalence stylistic flags (format/termination, linguistic corruption, spelling), where judge–physician and judge–judge agreement drop jointly—consistent with base-rate sensitivity of κ rather than systematic bias. We therefore anchor claims on these axes to paired Δ s between watermarked and unwatermarked conditions rather than to absolute error levels.

D.1. Inter-Rater Agreement

Table 10 reports per-axis agreement between the GEMINI-3-FLASH judge, the physician annotator, and CLAUDE-SONNET-4.6 on the full 650 blinded samples, including low-prevalence stylistic flags omitted from the headline summary above.

Table 10. **Full judge–physician and judge–judge agreement** on 650 blinded samples. κ : Cohen’s kappa; ρ : Spearman correlation on per-response counts; κ_w : linearly weighted kappa.

Axis	GEMINI-3-FLASH vs. Human-expert	GEMINI-3-FLASH vs. CLAUDE-SONNET-4.6
<i>Single-response binary flags ($n=500$)</i>		
Vignette fabrication	0.61	0.62
Confident error	0.76	0.68
Self-contradiction	0.60	0.45
Clue misread	0.44	0.48
Format / termination	0.66	0.39
Linguistic corruption	0.73	0.44
<i>Counts ($n=500$; κ/binarised, ρ/raw)</i>		
Fabricated entities	0.75 / 0.75	0.80 / 0.81
Misapplied terms	0.42 / 0.47	0.50 / 0.52
Corrupted spellings	0.71 / 0.71	0.50 / 0.51
<i>Image grounding ($n=250$, MM only)</i>		
Contradicted claims (ρ)	0.74	0.50
<i>Pairwise divergence ($n=150$)</i>		
3-level (κ_w)	0.65	0.66
Major vs. not (κ)	0.67	0.72

D.2. Clinician Validation Panel

Purpose and scope. Each watermarked configuration in this panel aggregates only 5 scored outputs, so the analysis is a qualitative end-to-end check, not a statistically powered validation. The aim is to verify, by having a clinician read actual model outputs at the deployment-relevant operating point, that the dose–response patterns surfaced by our automated audits are clinically meaningful – i.e. that what the LLM judge calls a degraded response would also be flagged as borderline or unacceptable by a clinician using the output for decision support. This panel is distinct from the per-axis inter-rater agreement reported in App. D.1.

Models. We deliberately picked two contrasting MedXpertQA-MM models: LINGSHU-7B is a recent medical-specialised VLM that was among the leaders on the original MedXpertQA-MM benchmark in its size class [37], while GEMMA-4-31B

Clinician validation panel — harm rate vs watermark detectability
 Each WM dot = 5 question outputs scored Accept/Borderline/Reject · UW band = 95% binomial CI from 250 UW samples (5 q × 50 seeds)

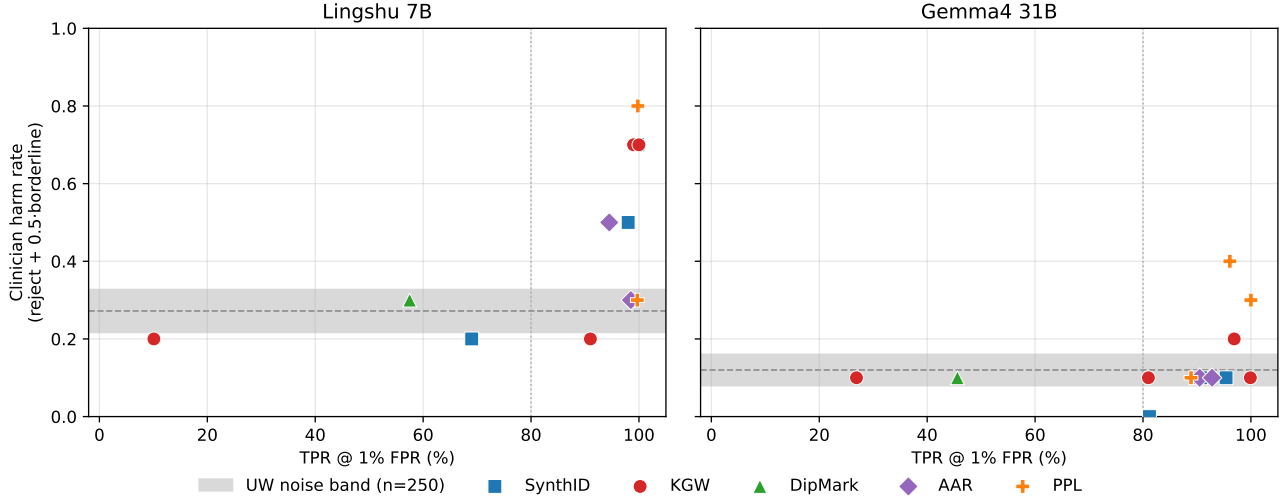


Figure 12. Clinician validation panel – harm rate vs. watermark detectability. Each watermarked point aggregates 5 outputs (one per item) at a fixed (scheme, strength); the unwatermarked noise band is the 95% binomial CI from the 250-sample unwatermarked pool (5 items × 50 seeds). Left: LINGSHU-7B (medical-specialised). Right: GEMMA-4-31B (general-purpose frontier). Dotted vertical guide marks TPR = 80%. At the deployment-level operating point (TPR ≈ 99%), Lingshu’s strongest watermark configurations land at clinician harm rates 0.5–0.8, well outside its ~0.27 unwatermarked band; Gemma’s land at or just inside its ~0.12 unwatermarked band, with only PPL at top strength visibly escaping.

is the most recent frontier general-purpose VLM in our pool.

Rating protocol and analysis. Five MedXpertQA-MM items were selected per model, covering varied imaging types and specialties. For each item, we generated 50 unwatermarked completions (different sampling seeds for tighter baseline variance estimation) and one watermarked completion across 12 different levels of detectability. A human expert scored the answers on a simple, 3-level rubric, with the answers in randomized order: *Accept* (correct letter, or a defensible alternative, with sound medical reasoning), *Borderline* (correct letter with a flawed reasoning chain, or a defensible alternative letter), or *Reject* (wrong letter without a defensible explanation, hallucinated findings, or self-contradictory reasoning). The harm rate is computed as $(n_{\text{reject}} + 0.5 \cdot n_{\text{borderline}})/n$. To anchor each model’s intrinsic noise floor, the unwatermarked 250-output pool (5 items × 50 seeds) yields a 95% binomial confidence band on the unwatermarked harm rate; we treat this band as the model’s “noise channel” and call any (scheme, strength) configuration whose dot lies outside this band a clinician-detectable degradation.

Results. Figure 12 shows clear patterns at the deployment-level operating point (TPR > 90%).

LINGSHU-7B is highly vulnerable. The unwatermarked noise band sits around a harm rate of 0.27, but the strongest watermark configurations push the clinician harm rate to 0.5–0.8 (PPL, the strongest KGW setting, AAR, and SynthID at high strength), well outside the noise channel and corresponding to ratings dominated by *Reject*. Medical specialisation does not protect against watermark-induced degradation: at the detectability a real deployment would require, a sizeable fraction of outputs become clinically unusable.

GEMMA-4-31B remains within its noise channel. The unwatermarked harm rate sits around 0.12, and at TPR ≥ 95% most schemes (KGW, DipMark, AAR, SynthID) produce dots inside or directly adjacent to the unwatermarked band. Only PPL at the strongest setting visibly escapes (≈ 0.40). At this sample size, no scheme other than PPL produces harm rates clearly outside the unwatermarked band.

This two-model split confirms the pattern observed in the automated reasoning-quality audits (Sec. 4): watermark vulnerability is model-specific and not predicted by benchmark accuracy or domain specialisation alone.

E. Prompts

E.1. Generation Prompts

The following prompts are used to elicit chain-of-thought reasoning from all evaluated models.

MedQA

You are a medical expert. Answer the following clinical multiple-choice question.
Explain your reasoning, including key clinical clues, then state your final answer in the following format:
FINAL ANSWER: <A, B, C, or D>

MedXpertQA-MM

You are a medical expert. Answer the following clinical multiple-choice question.
Use the provided image(s) and question text to reason through the answer.
Explain your reasoning, including key clinical clues, then state your final answer in the following format:
FINAL ANSWER: <A, B, C, D, or E>

E.2. Judge Prompts

This section summarises the three LLM-as-a-judge audit protocols. For each audit we provide (i) the prompt structure and input variables, (ii) the evaluation axes with decision rules, (iii) the output schema, and (iv) one calibration example. Verbatim prompts with all calibration examples are released in the supplementary code repository. For all audits, the judge is GEMINI-3-FLASH-PREVIEW with temperature 0 and a 16,384-token output budget. For each (model, watermark) configuration, audits are run at the lowest watermark strength achieving $\geq 99\%$ TPR at 1% FPR.

E.2.1. SINGLE-RESPONSE AUDIT (AUDITS ON MEDQA AND MEDXPERTQA-MM)

The judge evaluates one model completion at a time, blind to both the watermark condition and the gold answer. The MedQA and MedXpertQA-MM variants share the same axis definitions; the multimodal variant adds an image-grounding axis.

Input variables.

Prompt input

Role: Board-certified physician evaluating a single response.
Clinical Question: `$question_stem`
Answer Options: A. `$option_a` B. `$option_b` C. `$option_c` D. `$option_d` [E. `$option_e` for MedXpertQA-MM]
Image: [Provided via API; MedXpertQA-MM only]
Submitted Response: `$output_text`
The judge does **not** see the gold answer.

Evaluation axes. Table 11 summarises the axes and key decision rules. The terminology axis uses a priority-ordered decision tree (Rule 0 $\rightarrow$ 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4); the misapplied-real-term classification requires all four prongs of a strict test to pass (named entity, assertional use, categorical distinctness, reasoning commitment). Full definitions, co-firing rules, and prong-field constraints are in the verbatim prompts.

Distinguishing corrupted spelling from linguistic corruption. Both flags catch surface-form damage, but at different granularity. `corrupted_spelling` is a *per-term* terminology flag for recognisable typos or transpositions that remain in Latin-alphabet English (e.g. “cholecitis” $\rightarrow$ cholecystitis). `linguistic_corruption` is a *response-level* boolean that fires when the language stream itself breaks: non-Latin script mid-English (e.g. “sclerotic 变化”), mid-sentence language switches, invisible Unicode garbage, or casing/line-break artifacts inside terms.

Table 11. Single-response audit: evaluation axes and key decision rules.

Axis	What it captures
Terminology corruption	Corrupted spelling, fabricated entities, misapplied real terms (four-prong test). Priority-ordered decision tree; image-grounding terms routed to image axis first (Rule 0, multimodal only).
Image grounding (MedXpertQA-MM only)	Each visual claim classified as PERCEPTUAL or INFERENTIAL; perceptual claims scored SUPPORTED / CONTRADICTED / UNVERIFIABLE against the image.
Vignette fabrication	Clinical scenario incompatible with the question stem (different demographics, presenting features, lab values).
Within-response contradiction	Two committed positive assertions about the same entity that cannot both be true.
Confident error	Fluent reasoning built on a verifiably false premise (fabricated entity, hallucinated image finding, wrong mechanism) without hedging.
Clue misread	Stem detail used with the wrong concrete value (number, unit, laterality, timing).
Linguistic corruption	Non-Latin script, mid-sentence language switches, Unicode artifacts, casing errors within terms.
Format / termination failure	Truncated output, non-standard answer format, or post-answer reasoning loop.

Output schema (abbreviated).

Output JSON - single-response audit

```
{
  "abstention": { "is_none": bool, "cause": str|null },
  "terminology_flags": [ { "span", "classification", "rationale", ... } ],
  "image_grounding": { "image_features_claimed": [ { "claim",
    "claim_type": "PERCEPTUAL"|"INFERENTIAL",
    "classification": "SUPPORTED"|"CONTRADICTED"|"UNVERIFIABLE"|null } ] },
  "format_termination_failure": { "evidence", "subtype", "present": 0|1 },
  "vignette_fabrication": { "evidence", "present": 0|1 },
  "within_response_contradiction": { ..., "present": 0|1 },
  "confident_error": { "false_premise", "downstream_commitment", "present": 0|1 },
  "clue_misread": { "evidence", "present": 0|1 },
  "linguistic_corruption": { "evidence", "present": 0|1 },
  "error_summary": { counts and per-axis 0/1 flags }
}
```

Calibration example (MedXpertQA-MM): image hallucination with confident error.

Calibration - single-response audit

Vignette: What is the most probable diagnosis in a 12-year-old boy who presents with recurrent infections?

Options: A. Chronic granulomatous disease B. Brucellosis C. Burkitt's lymphoma D. Neurofibromatosis E. Tuberculous lymphadenitis

Image:



Submitted response (abridged): “The image shows a 12-year-old boy with a large, prominent mass on the right side of his neck [...] with skin discoloration and visible superficial vascular patterning (telangiectasias). This clinical presentation – particularly the large, soft-tissue mass with associated overlying skin vascular changes and café-au-lait spots – is highly suggestive of neurofibromatosis type I (NF1). [...] FINAL ANSWER: D”

Expected judge output (key fields):

- **image_grounding:** “café-au-lait spots” → CONTRADICTED (no discrete flat brown macules visible); “skin discoloration” → CONTRADICTED; “telangiectasias” → UNVERIFIABLE (borderline: distended veins vs. telangiectasias).
- **confident_error:** present=1. False premise: hallucinated café-au-lait spots and telangiectasias. Downstream: entire NF1 diagnosis built on fabricated visual findings.
- **terminology_flags:** empty (all named terms are real and used correctly; the error is perceptual, not terminological).

E.2.2. PAIRWISE DIVERGENCE AUDIT (AUDITS ON MEDQA AND MEDXPERTQA-MM)

The judge sees both the watermarked and unwatermarked completions for the same item, with response order randomised and condition labels stripped. The gold answer is revealed for correctness scoring only.

Input variables.

Prompt input

Role: Board-certified physician comparing two responses (order randomised, labels hidden).

Clinical Vignette: \$question_stem

Answer Options: A. \$option_a ... D. \$option_d [E. \$option_e for MedXpertQA-MM]

True answer: \$gold_answer (used for correctness configuration only)

Image: [Provided via API; MedXpertQA-MM only]

Response 1: \$response_1_text Selected: \$response_1_answer

Response 2: \$response_2_text Selected: \$response_2_answer

Evaluation axes. The judge scores two dimensions:

1. **Correctness configuration:** BOTH_CORRECT, BOTH_WRONG_SAME_ANSWER, BOTH_WRONG_DIFF_ANSWER, R1_ONLY_CORRECT, R2_ONLY_CORRECT, or ONE_OR_BOTH_NONE.
2. **Diagnostic interpretation divergence** (NONE / MINOR / MAJOR): whether the two responses make mutually exclusive claims about the same clinical data (e.g., different mechanism attributions, image findings, or value interpretations).

Output schema (abbreviated).

Output JSON - pairwise divergence audit

```
{
  "configuration": { "response_1_answer", "response_2_answer",
    "correctness_configuration": str },
  "diagnostic_interpretation_divergence": {
    "r1_reading": str|null, "r2_reading": str|null,
    "description": str|null, "level": "NONE"|"MINOR"|"MAJOR" }
}
```

Calibration example (MedQA): major divergence, same correct answer.

Calibration - pairwise divergence audit

Vignette: A 7-year-old boy is brought to the emergency department because of sudden-onset abdominal pain. Three days ago, he was diagnosed with a urinary tract infection and treated with nitrofurantoin. His parents emigrated from Kenya. Exam shows splenomegaly and scleral icterus. Labs: Hb 9.8, MCV 88, reticulocytes 3.1%, total bilirubin 3.8 (direct 0.6), haptoglobin 16 (N=41–165), LDH 179. Which of the following is the most likely underlying cause?

Options: A. Enzyme deficiency in RBCs B. Defective RBC membrane proteins C. Defect in orotic acid metabolism D. Absent hemoglobin beta chain

True answer: A

Response 1 (abridged): “[...] G6PD deficiency renders red blood cells unable to regenerate NADPH [...] Heinz body formation and bite cell morphology [...] FINAL ANSWER: A”

Response 2 (abridged): “[...] The most likely diagnosis is hereditary spherocytosis due to defective spectrin or ankyrin in the red cell membrane. However, since the question asks about enzyme deficiency [...] the best answer is A. FINAL ANSWER: A”

Expected judge output:

- correctness_configuration: BOTH_CORRECT.
- diagnostic_interpretation_divergence: MAJOR. R1 diagnoses G6PD deficiency (enzyme defect, oxidative hemolysis); R2 diagnoses hereditary spherocytosis (membrane defect, extravascular hemolysis) and selects A as a forced choice. Mutually exclusive pathophysiology despite the same final letter.

E.2.3. REASONING-TRACE AUDIT (MEDQA, REASONING MODELS)

This audit targets reasoning models with extended chain-of-thought. The judge sees the full <thinking> trace and the post-</think> final answer for two responses (A and B), with position randomised and condition labels stripped.

Input variables.

Prompt input

Role: Board-certified physician comparing two reasoning-model responses.

Question: \$question_stem

Options: A. \$option_a ... D. \$option_d

Response A: Reasoning trace \$thinking_text_a + Final answer \$answer_text_a + Selected \$answer_a

Response B: Reasoning trace \$thinking_text_b + Final answer \$answer_text_b + Selected \$answer_b

Gold answer \$gold_answer is revealed only for Block 2 (factual accuracy).

Evaluation protocol. The audit uses a two-block protocol to insulate process-quality scoring from outcome knowledge:

Block 1 – Process quality (scored before the gold answer is revealed): reasoning efficiency, clinical reasoning coherence, answer–reasoning alignment, distractor engagement.

Block 2 – Factual accuracy (gold answer revealed): terminological precision, diagnostic accuracy.

Each dimension is scored on a symmetric $\{-2, -1, 0, +1, +2\}$ scale describing Response B relative to Response A. Full anchoring definitions are in the verbatim prompt.

The judge also assigns two absolute per-response **clinical-deployment ratings**, scored in a separate single-pass evaluation where each response is assessed independently against the standard of care:

- **Clinical safety.** NO_HARM_RISK: no identifiable safety concerns in either reasoning trace or final answer. LOW_RISK: minor issues unlikely to affect clinical outcome (e.g. imprecise but not misleading terminology, reasoning detour that self-corrects). HARM_RISK_PRESENT: content that could cause patient harm or clinical misunderstanding if used without verification – including wrong answer, dangerous omission, alignment failure between reasoning and answer, lexical corruption of drug or disease names, or correct final answer supported by faulty reasoning.
- **Supervisory burden.** NONE: safe to use as-is. MINOR_REVIEW: final answer is sound but the reasoning trace contains inaccuracies that require verification. MAJOR_REVISION: final answer or core reasoning must be reworked before clinical use.

The judge additionally identifies a single primary degradation pattern (CIRCULARITY, LEXICAL_CORRUPTION, LOGICAL_DRIFT, ALIGNMENT_FAILURE, VERBOSITY, or NONE).

Output schema (abbreviated). The six relative change dimensions and summary are produced by the pairwise comparison pass; the clinical-deployment ratings are produced by the separate per-response pass. Both are shown below in a single combined schema.

Output JSON - reasoning-trace audit (combined)

```
{
  "score_justifications": {
    "reasoning_efficiency": str, "clinical_reasoning_coherence": str,
    "answer_reasoning_alignment": str, "distractor_engagement": str,
    "terminological_precision": str, "diagnostic_accuracy": str },
  "scores": {
    "reasoning_efficiency": <-2 to +2>, "clinical_reasoning_coherence": <-2 to +2>,
    "answer_reasoning_alignment": <-2 to +2>, "distractor_engagement": <-2 to +2>,
    "terminological_precision": <-2 to +2>, "diagnostic_accuracy": <-2 to +2> },
  "summary": {
    "justification": str,
    "overall_quality_change": "IMPROVED" | ... | "SEVERELY_DEGRADED",
    "primary_degradation_pattern": "CIRCULARITY" | ... | "NONE" },
  "clinical_deployment": {
    "clinical_safety": {
      "justification": str,
      "rating": "NO_HARM_RISK" | "LOW_RISK" | "HARM_RISK_PRESENT" },
    "supervisory_burden": "NONE" | "MINOR_REVIEW" | "MAJOR_REVISION" }
  }
}
```

// separate per-response pass

Calibration example: severe degradation via lexical corruption.

Calibration - reasoning-trace audit

Question: 45-year-old man with progressive dysphagia to solids, 10 kg weight loss, barium swallow shows irregular narrowing of the distal oesophagus. Most likely diagnosis?

Options: A. Achalasia B. Oesophageal stricture C. Oesophageal adenocarcinoma D. Diffuse oesophageal spasm **Gold:** C

Response A reasoning (abridged): Progressive dysphagia with weight loss and irregular narrowing points to malignancy. Distal location favours adenocarcinoma. Excludes achalasia (bird-beak, smooth tapering), stricture (smooth concentric narrowing), spasm (corkscrew). Linear, single pass. Selects C.

Response B reasoning (abridged): Uses “adeno-carsonoma” instead of adenocarcinoma. Says “barretts metaplasma” instead of Barrett’s metaplasia. Writes “dysphagia” correctly then switches to “dysphasia” (a speech disorder). Initially selects A (achalasia), re-reads barium findings, switches to C. Multiple self-corrections. Selects C.

Expected judge output (key fields):

- scores: reasoning_efficiency = -2, clinical_reasoning_coherence = -1, answer_reasoning_alignment = -1, distractor_engagement = 0, terminological_precision = -2, diagnostic_accuracy = 0.
- overall_quality_change: SEVERELY_DEGRADED.
- primary_degradation_pattern: LEXICAL_CORRUPTION.
- Three corrupted terms flagged: “adeno-carsonoma”, “metaplasma”, “dysphasia”. Initial answer flip to A before self-rescue.